Strategy as Ontology: From Metaphor to Mechanism in Strategic Thought

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ABSTRACT

Purpose

To establish the ontological foundations of strategy and advance a falsifiable, cross-domain framework that transmutes metaphor into mechanism.

Design / Methodology / Approach

This paper undertakes a conceptual synthesis rooted in critical realism, quantum probability, complexity science, and information theory. From a system-general definition of strategy—as a recursive inference–action loop that reduces epistemic entropy—it deduces eighteen universal laws. Canonical and contemporary literatures are interrogated to expose enduring conceptual lacunae.

Findings

(1) Strategy is ontologically construed as the means by which complex adaptive systems collapse probabilistic futures into realised advantage under conditions of entropy.

(2) The Quantum–Complex–Entropic–Adaptive (QCEA) paradigm formalises this view, revealing five structural deficiencies in extant frameworks: conceptual ambiguity, limited explanatory range, diagnostic opacity, theoretical sterility, and non-falsifiability.

(3) These are systematically resolved by the proposed eighteen laws, which render strategy observable, testable, and generalisable across military, corporate, biological, and digital domains.

Research Implications

Outlines a research programme for operationalising the laws through entropy-based metrics, recursive process-tracing, and agent-based simulation.

Practical Implications

Equips practitioners with tools to detect coherence drift, design adaptive architectures, and anticipate the inflection points of strategic timing.

Originality / Value

Recasts strategy as a measurable phenomenon of nature. It presents the first integrated, falsifiable ontology spanning micro-cognition to ecosystem dynamics—marking a decisive shift from heuristic bricolage to cumulative science.

Keywords: Strategy ontology; Critical realism; Management philosophy; Quantum cognition; Complex adaptive systems; Strategic coherence

INTRODUCTION

What distinguishes a strategic pivot from an operational flinch? When Meta rebranded and restructured itself around the metaverse, it was heralded as bold strategic foresight. When Nokia shifted too late toward smartphones, it was condemned as strategic failure. Yet both decisions involved uncertainty, resource reallocation, and directional bets under competitive pressure. By what criteria do we call one move "strategic," the other "mistaken"? What, in truth, do we mean by strategy?

This question is not rhetorical. It exposes a profound lacuna at the heart of management thought. "Strategy" saturates contemporary discourse—invoked by executives, generals, policy-makers, and algorithm designers alike—yet remains ontologically unmoored. The term suggests acuity and foresight, but inquiry into its essence yields contradiction, tautology, or evasion. As Joan Magretta once observed, "Everyone agrees strategy is important, but almost no one agrees on what it is."

Indeed, efforts to define it often generate more fog than light. Michael Porter framed strategy as the choice of unique positioning within an industry, protected by structural barriers. Henry Mintzberg identified five conflicting meanings: plan, ploy, pattern, position, and perspective. David Teece viewed strategy as the orchestration of dynamic capabilities amid shifting environments. Each framework offers insight. Yet none resolves the core ontological ambiguity: is strategy a noun or a verb, a plan or a process, a mindset or a mechanism?

This definitional opacity is not merely a conceptual inconvenience—it is a material failure. Institutions design strategy as though the world were static, their models linear, and uncertainty containable. But reality resists such naïveté. Environments are complex, interdependent, and unstable. SWOT matrices, logic models, and scenario planning tools presuppose stability where none exists. The result is what might be called strategic formalism: the appearance of rigour without ontological grounding.

By contrast, consider a slime mould—Physarum polycephalum—which, absent cognition or foresight, navigates a maze more efficiently than many executive teams. It does so not by design, but by recursive reinforcement and decay: strengthening viable paths, retracting others. Or consider AlphaGo, which astonished its creators by executing moves no human would predict, guided not by rules but by recursive inference and probabilistic adaptation. In both cases, what appears is not planning, but a system collapsing uncertainty into coherent action. Is this not strategy? If so, then human strategists may be the least reliable exemplars of their own field.

This paper arises from a simple but far-reaching provocation: What, across all domains, is strategy? It answers by proposing a universal ontological account. Drawing on systems realism, epistemic inference, information theory, and complexity science, it advances the Quantum-Complex-Entropic-Adaptive (QCEA) paradigm. This framework defines strategy as a recursive process by which adaptive systems reduce epistemic uncertainty and collapse superposed futures into realised, coherent configurations. Strategy, in this account, is not a plan but a pattern of inference and intervention—a probabilistic mechanism for converting possibility into advantage

under entropic constraints.

Three axioms underpin the paradigm:

1. Universality: Strategic phenomena manifest across evolution, cognition, and artificial systems—not merely in business or war.

2. Entropy: Strategy unfolds in environments defined by uncertainty, interdependence, and temporal decay.

3. Recursion: Strategic success derives not from singular design but from cycles of sensing, interpreting, cultivating coherence, and acting.

QCEA is offered not as a deterministic schema, but as a falsifiable hypothesis—a generalisable, mechanistic grammar of strategic action. It replaces metaphor with structure, narrative with mechanism, and intuition with testable form.

The paper proceeds as follows:

- Section II frames the definitional incoherence and its institutional consequences.
- Section III critiques the canonical corpus of strategy theory, highlighting its epistemic limits.
- Section IV outlines the abductive and interdisciplinary method.
- Section V maps the derivational logic underpinning the ontological account.
- Section VI presents the formal definition and its operational unpacking.
- Section VII derives eighteen universal laws for evaluating strategic coherence.
- Section VIII considers implications for theory, measurement, and practice, including boundary conditions and directions for future research.

The thesis is unambiguous: strategy merits the rigour of science, not the complacency of metaphor. To treat it otherwise is to mistake invocation for explanation—and in doing so, to perpetuate blindness where precision is most required.

II. THE PROBLEM: CONCEPT WITHOUT COHERENCE

Consider two companies in the video rental industry circa 2000: Blockbuster and Netflix. Both claimed to have clear strategies. Both employed competent executives, conducted market research, and made deliberate choices about resource allocation. Yet their conceptions of what strategy *is*—and thus what it demands—diverged fundamentally. That divergence proved fatal.

Blockbuster understood strategy through the lens of Michael Porter's theory of positional advantage. Their approach centred on:

- A dominant market footprint: 9,000 retail stores at peak
- Structural barriers: exclusive studio deals and high switching costs via late fees
- Economies of scale: overwhelming purchasing power and brand reach
- Market control: optimising revenue per store, per title, per square foot

Its CEO, John Antioco, declared in 2002: "Our strategy is to leverage our dominant market position and unmatched distribution network." This was not a failure of execution—it was a faithful application of *Porterian logic*. Blockbuster treated strategy as the preservation of competitive position, not as the cultivation of adaptive capacity. Its strategic reviews focused on defensive moats and optimisation—where to expand, how to reduce marginal costs—not on structural responsiveness to technological shifts.

When Netflix introduced DVD-by-mail, and later streaming, Blockbuster's leadership dismissed these as niche markets or premature innovations. These weren't failures of intelligence—they were consequences of a well-established strategic paradigm that equated advantage with structural entrenchment. By Porter's own criteria, Blockbuster was strategically

coherent—until the environment changed faster than its definitions could adapt.

Kodak offers a similar parable. The company invented digital photography but remained committed to its analogue business model. This decision wasn't irrational. It followed the logic of Ansoff's strategic planning framework, where growth is achieved by aligning product–market combinations. Kodak's product was film. Its market was global. Its strategy was to exploit synergy between innovation and market dominance—not to cannibalise the core for a speculative digital future.

Within this framework, Kodak optimised its R&D spend, invested in film innovation, and managed its portfolio through planned diversification. Digital initiatives were treated as adjacencies, not existential shifts. Kodak's failure wasn't myopia—it was ontological misdiagnosis. It saw the future coming, but interpreted it through a planning paradigm that presumed continuity rather than disruption.

In both cases, strategic failure resulted not from incoherence but from coherence with the *wrong definition*. These were not firms adrift—they were executing accepted frameworks with discipline and rigour. That their strategies failed reveals a more profound problem: the frameworks themselves mischaracterised the nature of strategy under conditions of environmental volatility.

This definitional ambiguity is neither isolated nor incidental. Netflix defined strategy as adaptive experimentation. Blockbuster defined it as positional advantage. Kodak defined it as planned growth. Each made rational decisions within its paradigm. Each exposed the fatal consequences of mistaking internally coherent definitions for ontologically sufficient ones.

Strategy, in its current usage, is an overloaded signifier. It means:

- Long-term planning (Ansoff)
- Competitive positioning (Porter)
- Emergent patterns (Mintzberg)
- Dynamic capabilities (Teece)
- Resource orchestration (Barney)
- Adaptive learning (March)

Each entails different behaviours, assumptions, and metrics. Yet all are called "strategy," and all are taught, invoked, and executed as though the term required no clarification. This is not pluralism—it is conceptual incoherence. A Tower of Babel where consensus vocabulary conceals ontological disagreement.

The implications extend beyond management. In military doctrine, strategy links political ends to military means. In artificial intelligence, it describes action selection under probabilistic constraint. In evolutionary biology, it denotes behavioural patterns that optimise fitness under ecological uncertainty. What unites these is not planning, but *emergence*. Not position, but *process*.

Philosopher W. B. Gallie (1956) referred to such terms as *essentially contested*—words that generate continual dispute because they reference underlying structures that remain ill-defined. Karl Weick (1995) went further: he called strategy a *sensemaking artefact*—a retrospective narrative imposed on uncertainty. In either case, the term lacks ontological discipline. It floats above the phenomena it purports to describe.

To stabilise it, we must ask not what strategy *ought* to be, but what strategy *is*. Across organisms, organisations, and algorithms—what is the mechanism that enables certain systems to collapse uncertainty into adaptive action?

To approach that question, we must briefly survey the dominant attempts to define strategy, and examine why each fails to deliver a generalisable account. Before doing so, we offer three empirical observations from radically different domains—biology, computation, and commercial collapse—that suggest strategy is not defined by intention, but by structure.

Biology. Cuttlefish camouflage, bonobo coalition-building, and damselfly deception all reveal

behaviours that align with strategic logic, despite lacking cognition. These organisms operate through recursive interaction with dynamic environments. They adapt not by design, but by iteration—strengthening viable behaviours, discarding ineffective ones.

Computation. AlphaGo's mastery of Go arose not from rule-following, but from recursive self-play. Large language models like GPTs collapse probabilistic distributions into coherent output through inference shaped by feedback. These are not metaphors for strategy. They *are* strategic processes—exhibiting coherence, recursion, and entropic constraint.

Business. Kodak invented digital photography but failed to let go of chemical revenue. Nokia saw the smartphone coming but misread the importance of ecosystems. Blockbuster had data, but not the will to cannibalise itself. These failures were not tactical—they were epistemological. The underlying mechanism for generating strategic coherence had decayed or was never fully developed.

These diverse systems—biological, artificial, organisational—all point to a shared pattern: recursive inference, feedback-aligned action, and emergent advantage under constraint. This is not narrative—it is mechanism.

To make this rigorous, we must replace vague invocation with testable structure. What follows is a brief comparative review of major definitional approaches in strategic theory, showing how each captures partial insight while failing to supply a unifying ontology.

TABLE I.

Thinker	Definition	Ontological Status	Key Limitations
Michael Porter	Positioning within industry structure	Static and exogenous	Ignores feedback, adaptation, and endogenous change
Henry Mintzberg	Strategy as plan, ploy, pattern, position, perspective	Descriptive, plural	Lacks coherence; collapses categories without ontological anchor
David Teece	Dynamic capabilities to sense, seize, and reconfigure	Emergent, partial	Omits inference mechanism; lacks criteria for failure

What these frameworks fail to supply is a mechanism capable of generating, maintaining, and diagnosing strategic coherence under uncertainty. Each names a facet. None explains the whole.

To progress, we require a formal, falsifiable definition grounded not in metaphor but in system behaviour—a definition that can be observed, measured, and interrogated across domains.

The next section turns to the canonical corpus of strategic thought—not to dismiss it, but to excavate what has been assumed, omitted, or mischaracterised in the journey from metaphor to mechanism.

III. FROM METAPHOR TO MECHANISM — THE UNFINISHED QUEST TO DEFINE STRATEGY

Strategy remains one of the most frequently invoked yet least defined concepts across fields. Executives, generals, algorithm designers, and evolutionary theorists all speak of "strategy" as if it were self-evident. Yet, as Magretta (2002) observed, "everyone agrees that strategy is important, but almost no one agrees on what it is." This epistemic fracture reverberates through decades of scholarship.

Mintzberg (1994) catalogued ten definitions, none of which command consensus. Rumelt (2011) decried strategy's evolution into a "catch-all term." Whittington (2001) warned of incommensurable paradigms, while Freedman (2013) noted that most accounts "start in the middle"—describing decisions, behaviours, or plans without clarifying what distinguishes strategic action ontologically.

This is not merely a philosophical grievance. The absence of ontological clarity obstructs

explanatory scope, empirical testability, and theoretical coherence. As Foss and Knudsen (2019) argue, strategy has grown more practically fragmented and conceptually vague over time. Strategy Science (2020–2024) reflects the trend: empirical richness, but continued foundational ambiguity.

To diagnose this void, we turn first to the canonical figures who approached the threshold of theory. Their contributions remain indispensable—but incomplete. We then turn to more recent additions and finally isolate the five critical absences preventing unification.

A. Porter: Structured Positioning in a Low-Entropy World

Michael Porter's work—Competitive Strategy (1980), Competitive Advantage (1985), and the later essay What is Strategy? (1996)—constitutes the most influential attempt to formalise business strategy through economic structure. Strategy, in Porter's formulation, is the deliberate positioning of the firm within its industry to achieve sustainable advantage through structural forces and internal consistency.

His tools—Five Forces, Generic Strategies, the Value Chain—brought clarity to competitive diagnostics. Managers now had a grammar for mapping suppliers, buyers, entrants, and rivals. His emphasis on fit—the interlocking coherence of activities—remains one of strategy's most operationalisable insights.

Yet Porter's framework assumes relative structural stability. Industry boundaries are presumed legible. Forces are treated as quasi-static. Advantage is durable because fit is hard to imitate. These assumptions falter in turbulent, reflexive environments.

Kodak exemplifies this failure. The company sustained advantage by optimising chemical film production and defending its retail channels—textbook Porterian logic. Yet the rise of digital photography—originally invented by Kodak itself—undermined the very structure its strategy presumed. Porter's models did not capture how strategic action could reflexively reshape the landscape. Nor did they accommodate information asymmetry, feedback loops, or path dependence.

QCEA addresses these gaps by treating industry structure not as a given, but as an emergent, entropic field. Strategy must collapse uncertainty within a shifting landscape, not merely fortify positions within a known one.

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B. Barney: The Resource-Based View (RBV) and Internal Sufficiency

Jay Barney's seminal paper (1991) shifted the locus of advantage from external structure to internal assets. Strategy, under the RBV, is the configuration of valuable, rare, inimitable, and non-substitutable (VRIN) resources that yield sustained performance.

This reorientation provided a counterweight to Porter's externalism. Firms now examined intangible assets—culture, patents, routines—as sources of differentiation.

Yet RBV often reifies what it seeks to explain. What makes a resource strategic? In practice, this is inferred ex post from success. There is no formal mechanism for determining how resources become valuable or under what environmental conditions their salience decays.

Nokia provides a sobering case. Its design teams and supply chains were once VRIN exemplars. Yet those same assets became liabilities when modular software ecosystems—not hardware differentiation—defined smartphone competition. RBV lacks an entropic lens for how environmental shifts erode resource value.

QCEA corrects this by embedding resources in dynamic feedback loops. Value is not static, but contingent on recursive coherence with a deforming landscape.

C. Teece: Dynamic Capabilities Without a Dynamic Ontology

Teece, Pisano, and Shuen (1997) extended RBV to environments of change. They defined dynamic capabilities as the firm's ability to integrate, build, and reconfigure competences in response to shifts. This was a pivotal turn. For the first time, strategy theory spoke of strategic adaptability as the centrepiece of advantage.

Yet the theory remains under-specified. What triggers capability reconfiguration? How are sensing, seizing, and transforming structured across time? Most accounts treat dynamic capabilities as black boxes: they exist if the firm adapts.

Intel illustrates the cost of this opacity. Despite sensing mobile disruption, its sunk investments and internal frictions impaired timely transformation. Its capabilities were dynamic in principle but inert in practice. What's missing is a mechanistic model of recursive inference that links signal detection, decision velocity, and systemic alignment.

QCEA renders this explicit. Adaptation is not merely reactive; it is a recursive cycle of strategic inference, in which the system narrows uncertainty by aligning knowledge, intention, and capability across temporal lags.

D. Clausewitz, von Moltke, and the Classical Turn Toward Friction

Clausewitz (1832) remains strategy's literary progenitor. His concept of **friction** captures the entropy of real-world execution: the gap between plans and reality. His dialectical treatment of war foreshadows complexity theory, acknowledging the fog, chance, and feedback inherent in conflict.

Yet Clausewitz offers metaphor, not mechanism. His famous "centre of gravity" provides no diagnostic model for inference. His fog is described, but not pierced. The strategist is left without an apparatus to collapse uncertainty or cultivate coherence.

Von Moltke the Elder pushed practice forward with Auftragstaktik—a decentralised doctrine allowing field units to adapt under mission command. Strachan (2007) and van Creveld (1985) interpret this as the birth of distributed cognition in strategy. Yet even this remains an operational doctrine without ontological scaffolding. It lacks a theory of how distributed agents sustain coherence across recursive time frames and feedback asymmetries.

QCEA integrates these insights into a structural theory of coherence under friction: entropy is not just acknowledged, but modelled as the field within which strategy operates.

E. Sun Tzu and the Seduction of Indirectness

Sun Tzu's Art of War weaves timeless themes—indirection, deception, timing, terrain—into a philosophical register. Sawyer (1994) and Griffith (1963) reveal its influence on cognitive warfare and perception shaping.

But Sun Tzu presumes an ideal strategist—a sage who discerns intent and intervenes flawlessly. The text lacks a model of inference under ambiguity, or of learning under emergence. It offers proverbs, not principles.

Modern equivalents abound. Huawei, for instance, employs Sun Tzu's logic to shape perception in international markets. But when environments turn reflexive—as with sanctions or trust collapse—proverbial guidance gives way to the need for diagnostic scaffolding.

QCEA retains the strategic virtue of indirection but embeds it in a formal system of recursive feedback, inference, and coherence evaluation..

F. The Rest: Fragments Without Foundation

Beyond these five, strategy's canon sprawls across contending epistemes — each illuminating, none unifying.

Henry Mintzberg's prolific corpus — The Rise and Fall of Strategic Planning (1994), Strategy Safari (with Ahlstrand and Lampel 1998), Tracking Strategies (2007) — remains the canonical critique of formal, deliberate planning. Mintzberg's enduring contribution is to reveal strategy's emergent character: that patterns of action often coalesce without central design, and that real strategists must learn to interpret signals rather than dictate blueprints. His taxonomies — plan, pattern, position, perspective, ploy — enrich the strategic vocabulary. Yet Mintzberg stops at pluralism. His schools of thought become a classification system, not a mechanism. He provides no generalisable, testable model for when emergence outperforms design, or how a system processes weak signals into actionable inference without descending into drift. The rise and dissolution of Nokia demonstrates this gap. Nokia's engineers frequently generated bottom-up innovations — classic emergent strategy — but without a formal architecture for integrating this emergence with coherent, top-down direction, innovations fragmented and failed to coalesce into durable advantage (Doz and Kosonen 2010). Mintzberg gives us the insight that strategy can emerge — but not the systemic rules that distinguish productive emergence from incoherent drift.

Alfred Chandler's Strategy and Structure (1962) gave the field its foundational historical ballast: his empirical work showed how large US corporations adapted organisational form to fit strategic expansion. His insight that "structure follows strategy" remains an enduring slogan for the link between design choices and strategic direction. Yet Chandler never defines strategy ontologically; he describes its artefacts retrospectively. His work remains a descriptive inference, not a falsifiable mechanism for how structure and strategy co-evolve in real time. The US conglomerate boom of the 1960s–70s, which drew heavily on Chandler's logic, illustrates this. Firms over-generalised the structure-follows-strategy dictum, building sprawling multi-divisional structures without mechanisms to sustain strategic coherence in rapidly shifting markets. As Rumelt (1974) later showed, many conglomerates disintegrated under the weight of unrelated diversification and sluggish information flows — the limits of Chandler's linear causality, absent a systemic feedback architecture.

Igor Ansoff's trilogy — Corporate Strategy (1965), Strategic Management (1979), Implanting Strategic Management (1984) — pioneered the formalisation of strategy as structured choice under environmental turbulence. He gave us the ideas of environmental scanning, capability profiles, and strategic posture, opening the door to treating strategy as a formal managerial discipline. Yet Ansoff's core model remains top-down and deterministic. He assumes that fit is a rational function of forecasting, ignoring how recursive feedback and emergent surprises warp planned trajectories. The collapse of General Motors in the 2000s illustrates this gap. GM mastered environmental scanning and formal planning — hallmarks of Ansoffian design — but failed to adapt to emergent shifts in consumer preferences, Japanese competition, and volatile fuel prices. Lacking an adaptive feedback loop, GM's strategy became a bureaucratic artefact rather than a living system (Hamel and Välikangas 2003).

Gary Hamel and C.K. Prahalad, in Strategic Intent (1989), Competing for the Future (1994), and The Core Competence of the Corporation (1990), refocused the field on foresight, ambition, and embedded capability. Their insight — that firms should stretch beyond existing resource constraints and nurture core competences — remains foundational for modern corporate advantage. Yet their work hovers above mechanism. They galvanise ambition but stop short of showing how strategic intent is recursively cultivated and recalibrated when external realities clash with visionary stretch. Sony's decline in the 2000s is telling: celebrated for its creative core competences, Sony failed to maintain coherence as divisions duplicated efforts and cannibalised innovation pipelines. Without a system to realign competencies amid technological convergence, its once-integrated intent fragmented (Cusumano and Selby 1995).

James March's seminal works — Organizations (1958, with Simon), Decisions and Organizations (1988), and Exploration and Exploitation in Organizational Learning (1991) — reframe strategy as a temporal tension between search (exploration) and refinement (exploitation). His elegant model captures how organisations balance short-term returns with long-term renewal. Yet March's constructs remain stylised. He does not formalise how a system detects when to switch regimes, nor how it structures feedback to ensure learning does not drift into aimless experimentation or ossify into myopia. Kodak's push into digital in the 1990s demonstrates this limit. It explored digital imaging technologies but failed to exploit them profitably before rivals captured market share — the exploration–exploitation balance broke down without a coherent system for learning under uncertainty (Lucas and Goh 2009).

Ralph Stacey's Complexity and Creativity in Organizations (1996), Strategic Management and Organisational Dynamics (2000), and Tools and Techniques of Leadership and Management (2012) brought complexity theory into management discourse. He shows how nonlinearity, bounded rationality, and self-organisation shape strategic possibility. Yet Stacey's insights stay theoretical. He describes the conditions under which complexity arises but does not construct strategy as a processual engine for converting emergence into advantage. BP's Deepwater Horizon disaster exemplifies this: BP's senior managers understood complex risk but lacked the organisational structures to transform local weak signals about technical failures into coherent, system-wide action before the crisis cascaded (Hopkins, 2012).

Karl Weick, through The Social Psychology of Organizing (1979), Sensemaking in Organizations (1995), and Making Sense of the Organization (2001), reframes strategy as post-hoc narrative coherence — sensemaking that stabilises uncertainty retrospectively. Yet sensemaking lacks anticipatory power. It explains how actors justify action after the fact but does not formalise how inference generates proactive intervention in complex, fast-shifting contexts. NASA's Challenger disaster shows this vividly. Engineers and managers engaged in sensemaking that normalised anomalies instead of generating clear diagnostic thresholds for action — the retrospective coherence papered over unsafe drift (Vaughan, 1996).

Dave Snowden's Complex Acts of Knowing (2002), The New Dynamics of Strategy (2003 with Kurtz), and Cynefin typology (2007 with Boone) translate complexity into practice through contextual intelligibility: agents must sense which domain they inhabit and adjust accordingly. Yet Cynefin is diagnostic, not generative. It describes domains — obvious, complicated, complex, chaotic — but does not formalise how agents recursively shape domains through intervention. The UK government's COVID-19 response confusion illustrates this gap. While Cynefin informed policy design, the lack of a systemic loop for shifting domains in practice meant diagnostic frames often lagged behind viral emergence and public behaviour (Snowden 2021).

Rita McGrath's Discovery-Driven Planning (1995, with MacMillan), Transient Advantage (2013), and Seeing Around Corners (2019) highlight adaptiveness as pre-emptive disengagement — when advantage erodes, move on. She clarifies why durable advantage is increasingly rare in high-velocity markets. Yet her work orbits the firm rather than the phenomenon. She does not build a falsifiable, general theory for how systems navigate multiple waves of advantage under emergent co-evolution. Intel's struggle to shift from declining PC chips to mobile illustrates this. Despite clear signals, Intel's rigid sunk investments and path dependencies undermined transient adaptation; no systemic method bridged strategy across waves (Doron and Gawer 2018).

Kathleen Eisenhardt, with Brown, Martin, and others, advanced the microfoundations of dynamic capabilities. Her landmark studies — Speed and Strategic Choice (1989), Dynamic Capabilities and Strategic Management (2000), Simple Rules (2015) — show how cognition, routines, and heuristics enable adaptation in fast-moving environments. Yet despite empirical sophistication, Eisenhardt's microfoundations presume rather than interrogate the ontological basis of strategy. She shows how firms adapt but does not articulate what strategy is as a general, testable process across domains. Yahoo's rapid decline illustrates this: teams used improvisation and 'simple rules' but lacked a coherent system to aggregate these micro-acts into durable advantage amid platform disruption (Fiegerman 2016).

Together, these contributions form a constellation of insights — methodologically rich, epistemologically fragmented. They produce lenses, not theory; phenomena, not principles. Their frames enable strategy's discussion, but not its definition. None traverse the ontological bridge from artefact to essence. This is not a failure of imagination, but of foundational inquiry. The question of what strategy is — across domains, time, and system boundaries — remains unposed.

Taken together, these canonical contributions — from Clausewitz's friction to Porter's structural logic, from Mintzberg's emergence to Rumelt's kernel — reveal a clear pattern: each frames strategy usefully within bounded conditions, yet none resolves the fundamental question of what strategy is across contexts. Their partial insights highlight not a failure of intellect but an evolutionary gap: the field has lacked a unifying mechanism capable of turning these fragments into a coherent, testable system..

G. Contemporary Strategy Debates

Having exposed the conceptual absences in classical frameworks, it is now necessary to map the current frontier. In recent decades, strategy research has fractured into fertile but siloed subfields. Each exposes critical insights—on cognition, discourse, routines, or networks—but few offer a unifying ontological grammar. The QCEA paradigm does not discard these strands. It reframes and operationalises them under structural laws, enabling commensurability without collapsing difference.

Strategy-as-Practice (SAP) — Whittington's (2006) call for "taller" ontologies is answered directly by QCEA's recursive loop from micro-inference to systemic coherence. Jarzabkowski's (2005) ethnographic depth surfaces the lived complexity of strategising but lacks falsifiable structure. Vaara and Whittington's (2012) practitioner–practice–praxis triad finds a structural analogue in QCEA's tripartite form: inference (subjective forecasting), cultivation (capability and alignment), and execution (collapse into realised outcomes). Where SAP thickens narrative texture, QCEA renders it diagnostically precise.

David Grant (2003, 2008) adds a critical dimension by analysing strategy as discourse. His integration of Bourdieu's relational thinking with strategy-as-practice foregrounds how identity, agency, and power are co-constructed through language. Most salient to QCEA are Grant's findings on *ontological security*—how discourse simultaneously disciplines and emancipates managerial action. Laws 6–8 of the QCEA framework, which address epistemic drift and perceptual misalignment, converge with this insight: strategist identity is not neutral but discursively regulated, shaped by the need to maintain coherence under uncertainty. Grant's work reframes the strategist not merely as decision-maker, but as meaning-maker under entropic pressure.

Behavioural Strategy — Gavetti and Levinthal (2000) show how backward-looking cognition distorts forward-looking inference. Powell, Lovallo, and Fox (2011) trace how heuristics morph into cognitive traps. Kahneman (2011) generalises these fallacies into systemic biases—anchoring, availability, representativeness—but lacks a structural framework for overcoming them. QCEA's Laws 6–8 formalise these phenomena as measurable epistemic distortions. Strategic error is not idiosyncratic; it is inferential lag under entropy. Perception gaps, belief inertia, and coherence loss are not symptoms of poor leadership but structural vulnerabilities to be anticipated, measured, and corrected.

Dynamic Capability Microfoundations — The microfoundations turn (Felin, Foss & Ployhart, 2023) corrects the oversimplification of routine-based advantage. Teece, Peteraf, and Leih (2016) redefine dynamic capabilities as agility in shifting landscapes. Helfat and Martin (2015) locate their roots in managerial cognition. QCEA reframes this: cognition is necessary, but not sufficient. Strategy emerges only when cognition aligns with system-wide coherence. Knowledge utilisation, feedback latency, and inferential load are not psychological variables—they are levers in the entropy equation of strategic survival. Without structural embedding,

capabilities degrade into competence traps.

Ecosystem Strategy & Platform Economics — Adner's (2012) ecosystem architecture, Jacobides et al.'s (2018) value blueprint, and Parker et al.'s (2016) two-sided market logic describe the external interdependencies that now define competitive dynamics. McIntyre and Srinivasan (2017) show that Netflix's edge lies not in scale, but in coherent orchestration: data feedback loops, content calibration, and pipeline synchronisation form a distributed inference engine. QCEA generalises this as cultivation under Laws 11–13: ecosystems do not merely surround the firm—they co-evolve with it. Strategic advantage lies in how well context is sensed, shaped, and synchronised.

Synthesis — These contemporary strands are not reducible to a single paradigm, nor must they be. But their insights become structurally comparable only when nested within a coherent ontological grammar. QCEA provides that grammar. It does not overwrite ethnography with equation, or psychology with physics. Rather, it shows how inference, agency, and emergence can be modelled under entropic constraint—so that strategy becomes not just narratable, but testable.

Having now surveyed both the classical deficiencies and the frontiers of modern debate, the paper turns to the paradigm that seeks to resolve them: the Quantum-Complex-Entropic-Adaptive theory of strategy.

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H. The Five Recurring Absences: Literature-Supported Diagnosis

Despite the voluminous contributions to strategic thought, the field continues to orbit five recurring deficiencies—deficiencies that prevent strategy from maturing into a unified, testable theory of adaptive advantage. These absences are not merely conceptual lacunae; they carry profound empirical and methodological implications. Each is surfaced by thinkers already examined, and each is echoed—often with mounting urgency—across decades of literature in business, military science, philosophy of science, and organisational theory.

1. Conceptual Clarity

Strategy remains, in Gallie's (1956) terms, an "essentially contested concept"—a placeholder for insight rather than a clearly bounded phenomenon. From Clausewitz's friction to Sun Tzu's indirectness, and from Weick's sensemaking to Mintzberg's five Ps, scholars have provided evocative metaphors, but few have specified the structural properties that render an action or process "strategic" (Chia & Holt, 2006; Rumelt, Schendel & Teece, 1994). The definitional opacity persists despite widespread acknowledgement: Rumelt (2011) laments that strategy has become a "catch-all term," while Magretta (2002) observes that "almost no one agrees on what it is." Systematic reviews continue to confirm this ambiguity (Bailey et al., 2022; Whittington, 2001), with scholars disagreeing not only on the components of strategy, but on its fundamental ontology. The field thus suffers not from pluralism, but from conceptual evasion.

2. Explanatory Scope

Most strategic frameworks remain resolutely domain-bound. Porter's industrial logic, while powerful in markets with stable structures, does not generalise to adaptive ecosystems, biological systems, or distributed artificial agents (Siggelkow, 2001; Gavetti & Levinthal, 2004). Clausewitz's insight into warfare, while epistemically rich, remains inapplicable to digital platforms or ecological competition. Cynefin's typologies (Kurtz & Snowden, 2003), though influential in organisational storytelling, falter when transferred to biological emergence or machine strategy. The absence of a transferable model impairs interdisciplinary application—limiting insights from cyber strategy (Borghard & Lonergan, 2017), biomimetic innovation (Benyus, 2002; Vincent et al., 2006), or public-sector policy (Christensen et al., 2016). Without a system-general foundation, strategy fragments into local tools rather than cumulative theory.

3. Diagnostic Power

A third deficit lies in the field's reliance on retrospective narrative rather than predictive structure. Strategic coherence is often diagnosed post hoc—once failure has occurred or success has been mythologised (Langley, 1999; Rumelt, 2011). Even in works that foreground coherence—such as Rumelt's kernel or Porter's activity fit—there is little guidance on how to detect incoherence as it emerges. Scholars like Chia and Holt (2006), Eisenhardt and Martin (2000), and Stacey (1996) highlight this narrative bias. Sensemaking approaches, though rich in explanatory colour, fail to specify mechanisms of collapse or maladaptation in real time (Chelariu et al., 2002; Hodgkinson & Starbuck, 2008). The result is a diagnostic void: we recognise coherence only in hindsight, and its erosion only in autopsy.

4. Generativity

Fourth, strategy lacks generative power. Few frameworks yield modular, recombinable components that can be simulated, scaled, or applied in unfamiliar contexts. Rather than producing a family of mechanisms or design principles, the literature recycles typologies—lists of patterns, schools, or heuristics (Bettis et al., 2014; Burgelman et al., 2018). March's (1991) exploration/exploitation model remains a metaphorical axis, rarely operationalised into structural mechanisms. Recent work in circular economy, adaptive leadership, and digital transformation repeats the complaint: strategy lacks mechanisms for engineered emergence (Birkinshaw, 2017; Kohtamäki et al., 2021). Where it should enable new architectures of advantage, it yields recycled vocabulary.

5. Falsifiability

Finally, strategy remains largely unfalsifiable. Most propositions are retrospective, interpretative, or tautological—rarely offering hypotheses that can be empirically refuted (Eisenhardt & Zbaracki, 1992; Durand & Vaara, 2009). Huff (1990) and Suddaby et al. (2013) have argued that strategy studies rarely generate boundary-defining statements that permit robust testing or simulation. Even advances in dynamic capabilities or microfoundations struggle to yield formal, cumulative models (Makadok et al., 2018). As long as strategic claims remain flexible narratives rather than precise predictions, the field cannot evolve under the discipline of empirical science.

Meta-Critiques: Fragmentation and Conservatism

Two further themes deepen the malaise. First, epistemological fragmentation: the coexistence of incommensurable paradigms—rational choice, behavioural, institutional, complexity—without ontological reconciliation (Whittington, 2001; Johnson et al., 2007). Second, methodological conservatism: a heavy reliance on retrospective case studies, with limited uptake of simulation, formal modelling, or hybrid methodologies (Bettis et al., 2014; Suddaby et al., 2013). These meta-patterns preclude theory consolidation and suppress the emergence of falsifiable structure.

This framework surfaces not a void, but an evolutionary bottleneck: strategy has become a field of brilliant fragments, unmoored from system-generalisable theory. The next section introduces a paradigm designed to address each of these absences—a formal ontological framework grounded in entropy, coherence, and adaptive recursion.

H. Tables of Comparative Limits

TABLE II.

Framework	Definition	Strength	Blind Spot	What QCEA Adds	Criteria Lacked by Framework	Criteria Addressed by QCEA
Porter	Position within industry structure	Structural clarity and fit logic	Ignores reflexivity and path deformation	Strategy as recursive collapse of deforming structure	Explanatory Scope; Generativity; Falsifiability	Explanatory Scope; Generativity

Framework	Definition	Strength	Blind Spot	What QCEA Adds	Criteria Lacked by Framework	Criteria Addressed by QCEA
Barney	Possession of VRIN resources	Endogenous sources of advantage	No mechanism for value erosion or revaluation	Recursive resource calibration under informational entropy	Diagnostic Power; Generativity; Falsifiability	Generativity
Teece	Dynamic capabilities to adapt	Acknowledges turbulence and change	Underspecified triggers and update mechanisms	Structural inference loops for cultivation and reconfiguration	Diagnostic Power; Falsifiability	Diagnostic Power
Clausewitz	Action under friction and uncertainty	Recognises entropy and chance	No method for coherence restoration or feedback	Entropy as field; coherence as recursive system property	Conceptual Clarity; Diagnostic Power; Generativity	Conceptual Clarity; Diagnostic Power
Mintzberg	Pattern of action that emerges over time	Captures bottom-up emergence	No thresholds for coherence vs drift	Cultivated coherence via recursive selection and pruning	Conceptual Clarity; Generativity; Falsifiability	Generativity
Rumelt	Coherent linkage of diagnosis and action	Distinguishes good vs bad strategy	No mechanism for updating coherence dynamically	Diagnostic architecture for coherence under emergence	Generativity; Falsifiability	Diagnostic Power
Chandler	Structure follows strategy	Historic link between intent and form	Retrospective causality; no dynamic adjustment	Co-evolution of structure and intent in real time	Diagnostic Power; Generativity; Falsifiability	Diagnostic Power; Generativity
Ansoff	Structured fit with environment	Formalised planning under turbulence	Top-down; ignores recursive surprise	Recursive sensing and update architecture	Conceptual Clarity; Generativity; Falsifiability	Diagnostic Power; Generativity
Hamel & Prahalad	Strategic intent and core competence	Ambition and capability logic	No system for recalibrating intent amid change	Recursive cultivation of capability under coherence constraints	Generativity; Diagnostic Power; Falsifiability	Generativity
March	Exploration vs exploitation trade-off	Temporal framing of strategic tension	No rule for switching or feedback modulation	Entropic signals as regime-shifting triggers	Generativity; Falsifiability	Diagnostic Power; Generativity
Stacey	Complexity, nonlinearity, self-organisation	Rich complexity framing	No mechanism for converting emergence to strategy	Entropic boundary detection and coherence engineering	Conceptual Clarity; Diagnostic Power; Generativity; Falsifiability	Conceptual Clarity; Diagnostic Power; Generativity
Weick	Retrospective sensemaking	Interpretive insight into action	Post-hoc bias; lacks predictive structure	Prospective inference-action alignment under uncertainty	Falsifiability; Diagnostic Power; Generativity	Diagnostic Power
Snowden	Contextual typology (Cynefin domains)	Context-sensitive diagnostics	Static domain classification; no transition logic	Recursive domain shaping through entropic feedback	Generativity; Falsifiability	Generativity
McGrath	Strategy as transient advantage	Emphasises timing and reinvention	No system-level navigation across advantage waves	Entropic landscape mapping and option path design	Explanatory Scope; Diagnostic Power; Falsifiability	Explanatory Scope; Diagnostic Power
Eisenhardt	Microfoundations of adaptive routines	Fast-paced, evidence-based adaptation	Lacks generality; no system-level architecture	Modular inference-action scaffolding for systemic adaptation	Conceptual Clarity; Explanatory Scope; Falsifiability	Explanatory Scope; Generativity

TABLE III.

Contemporary Debate	Key Contributions	Mapped to QCEA Domain	QCEA Advance
Strategy-as-Practice	Praxis, micro-activity, situatedness (Whittington, Jarzabkowski)	Inference–Cultivation–Execution triad	Structural coherence and measurement across levels
Behavioural Strategy	Cognitive error, heuristics, judgment under uncertainty (Kahneman, Gavetti)	Perception–Reality Drift, Epistemic Misalignment	Falsifiable diagnostic laws of inferential error
Dynamic Capabilities	Update routines, adaptive capability (Teece, Felin)	Cultivation of adaptive coherence	Operationalisation of coherence under drift
Ecosystem Strategy	Co-evolution, platform interdependence (Adner, Jacobides)	Contextual Cultivation, Timing, Fit	Recursive alignment in multi-agent systems

TABLE IV.

Combined Frameworks	Conceptual Clarity	Explanatory Scope	Diagnostic Power	Generativity	Falsifiability	Fragmentation	Conservatism
All Classical Models Taken Together	✕	✕	✕	✕	✕	✓	✓
QCEA Framework	✓	✓	✓	✓	✓	✕	✕

The point is not to dismiss the canon, but to complete it. The thinkers above gave us invaluable tools: strategic insight, metaphorical richness, diagnostic lenses. But none succeeded in offering a unified, falsifiable theory of strategy as an adaptive system.

The next section introduces the QCEA paradigm as such a theory: one that formalises the recursive process by which agents collapse uncertainty into advantage within entropic fields.

It seeks not to replace the canon, but to explain what makes each fragment strategic, and where its boundary lies.

IV. PHILOSOPHICAL RIGOR: KEY ONTOLOGICAL STREAMS

If this paper seeks not merely to propose a framework but to reconstitute strategy as an ontological category, then its philosophical lineage must be stated with clarity, if not defiance. The Quantum-Complex-Entropic-Adaptive (QCEA) paradigm does not drift rootless in abstraction. It draws from, extends, and in some cases contests, four principal ontological traditions: process philosophy, pragmatist epistemology, relational ontology, and phenomenology of practice.

Process Philosophy — Most directly, QCEA aligns with process ontologies derived from Alfred North Whitehead and extended by Robert Chia and Robin Holt (2006) and Tor Hernes (2014). Here, becoming precedes being; entities are entangled sequences of prehended events. Chia and Holt render strategy not as a thing but as an emergent pattern of becoming. This resonates with the QCEA notion of "dancing landscapes," yet differs in orientation: while Chia foregrounds ontological flatness and indeterminacy, QCEA imposes a gradient—entropy. Kauffman's adaptive landscapes (1993), Van Valen's Red Queen hypothesis (1973), and the probabilistic contours of selective pressure introduce topological asymmetry. Where process theory celebrates flux, QCEA models its structure.

Andrew Pickering (2010), in *The Cybernetic Brain*, traces the historical convergence between cybernetics and ontological becoming. His account helps position QCEA as the heir not only to metaphysical traditions but to a lineage of adaptive control: Ashby (1956), Beer (1972), and von

Foerster (2003) echo in the QCEA emphasis on recursive feedback.

Tor Hernes' notion of "time-knots" (2014) offers yet another bridge: the QCEA laws concerning temporal coherence, sequence, and timing resonate with this framing. Yet, where Hernes maps temporality as a managerial artefact, QCEA derives it as a universal constraint—entropy renders time not optional but axial.

Pragmatist Ontologies — The QCEA paradigm also extends from, and departs from, the American pragmatist tradition. Dewey's notion of inquiry as problem-solving through experimental action under uncertainty (Crouter & Soares, 2017) anchors QCEA's inferential engine. Yet QCEA is not content with anti-foundationalism. Where Rorty (1989) urges irony and contingency, QCEA insists on falsifiability: not a rejection of truth claims, but a demand that they be testable under drift.

Hans Joas (1993) introduces a critical enrichment—creativity as the inflection point where habituated patterns collapse and new strategies emerge. This mirrors the QCEA law that coherence must renew through selective error correction, not stasis. Joas helps embed the laws of adaptation not in computation but in the lived tension between past stability and future possibility.

Relational Ontologies — The systemic ambition of the QCEA paradigm finds ontological grounding in relational theories of becoming. Niklas Luhmann's *Social Systems* (1995) reframes society not as a network of autonomous actors but as self-referential communication loops. Meaning emerges not from discrete entities, but from the structural coupling of operations—each recursively constituted by signal, feedback, and adaptation. QCEA mirrors this orientation: strategy is not the act of a sovereign agent, but the emergent product of inference-action loops embedded in and shaped by selective environments. Agency is not a property of the individual but an outcome of recursive participation in a distributed informational system.

This coupling logic extends into sociomaterial and digital ontologies. Sjöstrand (2022) and Reid & Jones (2021) recast digitality not as infrastructure, but as relation-in-motion: coherence arises from the dynamic calibration of signals, nodes, and constraints. In platform ecosystems like Netflix, coherence is not imposed but cultivated—through inferential synchrony, user-feedback loops, and the adaptive realignment of content and architecture. These domains enact strategy through entangled, reflexive flows, not command hierarchies. QCEA formalises this entanglement as structural coherence under drift—diagnosable, decaying, and reconfigurable.

Maurizio Ferraris deepens this architecture through his theory of *documentality* (2005, 2013). He argues that social objects do not arise from collective intentions alone, but through inscribed acts—records, contracts, plans—that anchor social reality. This insight reinforces QCEA's emphasis on strategic artefacts: dashboards, forecasts, memos, scenario plans. These are not ephemera but constitutive elements—anchors of coherence whose degradation signals entropic drift. Strategy, under QCEA, is not merely enacted but recorded; and what is recorded becomes real, if only provisionally.

Graham Harman's object-oriented ontology (2002, 2018) offers a final provocation. Strategic entities—organisations, ecosystems, doctrines—are not exhaustively knowable. They withdraw from total access, yet exert causal force through their manifestations. Recent organisational theory positions these as *entrepreneurial objects*: ontologically receding but operatively generative. QCEA aligns with this stance. A strategy is never fully visible in its artefacts or expressions. It is traceable through coherence, through recursive pattern, through the persistence of advantage under drift. Yet it remains more than its measures: a structural object whose causal efficacy exceeds its observable form.

Thus, QCEA's relational ontology binds digital dynamism, communicative coupling, documentational force, and object-oriented opacity into a unified explanatory grammar. Strategy, here, is a structurally entangled, recursively enacted, and causally potent object—at once emergent and enduring, visible in parts but irreducible in whole.

Phenomenological Grounding — Finally, phenomenological currents enter not as footnotes but as force. Strategy is lived before it is modelled. Schultz and Tsoukas (2011) introduce

temporality not as abstract clock-time, but as lived experience of emergence. Sandberg and Dall'Alba (2010) add that strategic judgement is not reducible to calculation—it is embodied inference under pressure.

Chia's empirical study of NHS reform (2008) exemplifies this: the recursive entanglement of belief, friction, and local adaptation finds expression in real-time decision experience.

Lisa Segal (2022) reinforces this phenomenological strand with existential-hermeneutic methods. Her reframing of project management as an existential praxis, drawing on Heidegger and Rorty, surfaces ontological disruptions as lived managerial experience. These disruptions mirror QCEA's treatment of inferential breakdown (Laws 6–8), where managers must reconstruct meaning under epistemic erosion.

Similarly, Cicmil (2021) co-develops an existential-hermeneutic lens for organisational change, highlighting the incoherence practitioners feel amid metric-driven reform. His critique of managerial instrumentalism supports QCEA's indictment of post-hoc rationalisations and aligns with its demand for real-time coherence diagnostics (Laws 11–12).

QCEA acknowledges the affective and subjective dimensions of strategy without collapsing into relativism. The laws are structural, but their manifestation is phenomenological. In strategy, as in existence, coherence is lived before it is calculated.

Synthesis — Each of these traditions grants the QCEA paradigm depth and discrimination. Process philosophy informs its recursive core; pragmatism infuses its epistemic realism; relational ontology scaffolds its systemic generality; phenomenology grounds it in human agency. Yet the QCEA framework remains distinct: it operationalises these streams under entropic constraint. It does not merely describe strategic becoming—it defines the necessary conditions under which becoming constitutes strategy.

The ontological fracture long buried beneath strategic theory is here exposed and re-stitched. What follows is not a pluralist gesture, but a generalisable structure—one that does not refuse contingency, but insists that even in drift, there are laws.

V. METHODOLOGICAL APPROACH

i. Philosophical Grounding: Realism Without Determinism

The Quantum-Complex-Entropic-Adaptive (QCEA) paradigm rests on a foundational ontological claim: strategy is real, recursive, and entropically constrained—but not determined. This position rejects both naïve empiricism and metaphysical idealism. It affirms scientific realism while absorbing the complexities introduced by quantum indeterminacy, informational entropy, and complex-adaptive emergence.

We adopt realism in the Bhaskarian sense: there exists a stratified reality independent of observation, but our knowledge of it is always inferential, provisional, and mediated (Bhaskar, 1975; Bunge, 2006). Strategic systems—markets, militaries, ecosystems—exhibit structural properties that endure beyond perception. Yet strategic knowledge is not access to truth, but an ongoing contest with lag, noise, and decay. As information ages, its epistemic clarity deteriorates. In strategy, knowledge is entropic.

To maintain explanatory fidelity across layers, QCEA integrates four distinct but complementary ontological domains:

Quantum Mechanics informs the probabilistic nature of inference and possibility—superpositions of futures that collapse only through action.

Complexity Theory explains emergence, feedback, and coherence across interacting agents and structures.

Information Theory quantifies uncertainty and formalises the decay of knowledge.

Epistemic Realism asserts the mind-independent existence of structures, even as they are perceived only through signal and inference.

This is not metaphysical fusion. It is stratified complementarity—each layer speaks to a distinct domain of strategic reality. Quantum mechanics maps aleatory futures; complexity models dynamic interdependence; information theory measures decay; realism secures the world's ontological resistance.

Yet realism alone cannot explain how structure and agency co-evolve across time. For this, QCEA draws on Margaret Archer's morphogenetic approach (1995, 1998). Archer rejects the conflation of structure and agency. Her analytical dualism separates the two conceptually to trace their causal interplay. In her morphogenetic cycle, systemic conditions (T1) shape the field of possible action; agents act within and against these constraints (T2–T3); and through this action, structures are either reproduced or transformed (T4).

QCEA maps directly onto this sequence. Law 1—The Information-Action Cycle—models the recursive passage from inherited strategic conditions to novel outcomes. Strategic agents operate under constraint, process decaying signals, and attempt coherent interventions. These interventions in turn modify the informational terrain, resetting the conditions for the next cycle. Archer's T1–T4 becomes, within QCEA, a closed feedback loop of inference, cultivation, and collapse.

Like Archer's framework, QCEA holds that novelty is not a threat to realism. It is its proof. Emergence is not chaos; it is structured transformation. But this transformation is only intelligible when temporality is explicit, constraints are traced, and recursive mechanisms are modelled.

Thus, QCEA affirms realism without determinism, agency without voluntarism, and structure without stasis. It is a layered ontology of strategic reality: recursive, entropic, and intelligible under drift.

ii. Standards for Interdisciplinary Coherence

The QCEA paradigm does not traffic in metaphor. It imposes discipline on interdisciplinary ambition. As Oswick et al. (2011) caution, theoretical integration often collapses into conceptual tourism—terms borrowed, not built; metaphors imported, but never operationalised. Nicolini (2009) identifies this slippage as a central failing of management theorising: the erosion of explanatory traction through linguistic allure. Bechtel and Abrahamsen (2005) warn that without structural correspondence between levels of explanation, multi-domain models devolve into incoherent bricolage.

To prevent such drift, QCEA enforces three architectural standards:

Functional Alignment — Concepts are included only if they perform a definable causal role within the orchestration of strategy. "Entropy," "superposition," and "collapse" are not rhetorical artefacts. "Entropy" captures epistemic uncertainty and decision decay; "superposition" denotes the provisional coexistence of competing strategic hypotheses; "collapse" formalises the moment inference is resolved into commitment. No term survives without explanatory labour.

Level Specificity — Ontological levels remain stratified. Quantum mechanics applies to probabilistic inference, not organisations. Complexity governs emergence in systems, not the behaviour of particles. Each theory remains within its domain, but is structurally aligned. Cross-level claims require formal mapping, not metaphorical drift. This stratification avoids what Bechtel and Abrahamsen (2005) identify as "vertical conflation"—the confusion of mechanistic explanation with metaphorical resonance.

Translation Integrity — Terms are imported with semantic fidelity. If redefined, the redefinition is explicit. "Collapse" does not refer to quantum decoherence, but to an epistemic act

of commitment. "Entropy" does not mirror thermodynamic disorder, but quantifies decision-space uncertainty. This rigour ensures the model remains intelligible, portable, and falsifiable.

To this, Paul Griseri (2002, 2013) adds a critical dimension: epistemological precision is not merely academic—it is ethicalGriseri contends that without ontological clarity, managerial knowledge becomes both normatively ungrounded and politically evasive. His insistence that falsifiability is a condition of responsible theory-building aligns precisely with QCEA's architectural intent. Each law is not a metaphorical gesture but a testable constraint—binding insight to structure, and structure to consequence.

These standards are not appendices. They are the load-bearing constraints of the paradigm. QCEA does not merely coordinate disciplines; it disciplines them. It resists the seduction of conceptual promiscuity and instead forges an integrated theory that is structurally robust, ontologically explicit, and ethically defensible.

A. Iterative Qualitative Synthesis and Abductive Reasoning

The Quantum-Complex-Entropic-Adaptive (QCEA) paradigm emerged from a systematic, iterative process best described methodologically as a rigorous qualitative synthesis, enriched by continuous abductive reasoning. Qualitative synthesis entails a structured integration of insights across diverse theoretical literatures, empirical experiences, and critical reflections (Noblit and Hare 1988; Dixon-Woods et al. 2005). Abductive reasoning, first articulated by Peirce (1903) and developed extensively in qualitative research by Dubois and Gadde (2002), describes an iterative inferential approach where theoretical frameworks and empirical evidence inform each other reciprocally, progressively refining theoretical insights toward explanatory coherence.

This iterative, abductive approach unfolded through sustained scholarly engagement with strategic literature, empirical practice, and ongoing conceptual refinement. Over more than a decade, canonical strategic thinkers—including Porter (1980, 1985, 1996), Mintzberg (1987, 1994), Rumelt (2011), Freedman (2013), Clausewitz (1832), Sun Tzu (Griffith 1963; Sawyer 1994), Hamel and Prahalad (1989, 1990, 1994), Chandler (1962, 1977), Ansoff (1965, 1979, 1984), Stacey (1996, 2000), Snowden (2002, 2007), Weick (1979, 1995), McGrath (1995, 2013), and Eisenhardt (1989, 2000)—were revisited multiple times, not merely as academic references, but as lenses tested rigorously against strategic problems encountered in organisational leadership and strategic advisory practice.

The methodological orientation of this inquiry partakes of the tradition of systematic conceptual analysis and synthetic philosophy that has yielded many of the most architectonic frameworks in the history of thought. As with Kant's decade-long refinement of the *Critique of Pure Reason* between its first (1781) and second (1787) editions, or Heidegger's two-decade gestation of *Being and Time* through iterative lectures and philosophical recalibration, this work is the product of sustained dialectical labour rather than empirical aggregation.

The approach conjoins Strawson's (1959) *descriptive metaphysics*—the effort to disclose the most general features of our conceptual scheme—with Sellars' (1963) notion of the *synoptic vision*: the bringing into coherence of insights drawn from formally distinct domains. In more contemporary terms, it resembles the conceptual persistence evidenced in Harman's (2018) articulation of object-oriented ontology, or DeLanda's (2006) assemblage theory, which fuses Deleuzian metaphysics with the rigour of complexity science. In each case, conceptual architecture emerges not from data, but from an exacting synthesis of abstraction and applicability.

The method deployed here is most precisely described as *recursive conceptual synthesis*. Practical entanglements in the domain of strategy repeatedly compelled a return to theoretical first principles. These engagements refined the conceptual schema, which, in turn, confronted further empirical tensions—an iterative dialectic not unlike the generative cycles of Whitehead's process philosophy, itself shaped by the interplay of mathematical formalism and metaphysical inquiry. The resulting QCEA framework thus crystallised through a decade of oscillation between problem

and principle, collapsing surface volatility into a deeper structural ontology, where quantum probability, informational entropy, and complex adaptivity yield a coherent, cross-domain theory of strategy.

The methodological objective was not mere theoretical comparison; rather, it aimed to rigorously assess each canonical contribution's conceptual completeness, empirical coherence, explanatory power, and limitations. Through repeated practical application, this process systematically revealed which aspects of strategy each canonical thinker illuminated and which elements remained unresolved. Porter's frameworks clarified Formulation, Resource Allocation, and Imposing Coherence under stable conditions, while Mintzberg's work foregrounded Recursion and the Complex-Adaptive nature of emergent strategy. Rumelt strengthened understanding of Knowledge-Driven Diagnosis and Inferential Decision-Making, and Freedman underscored the Continuous Creation of Advantage under Selective Pressure. Clausewitz and von Moltke highlighted Environmental Friction, Emergent Landscapes, and Recursive Adaptation in conflict contexts. Sun Tzu's emphasis on Indirectness, deception, and timing foreshadowed the amplification of favourable outcomes. Hamel and Prahalad clarified Strategic Intent and Capability Cultivation, while Chandler and Ansoff demonstrated how Formulation, Cultivation, and Execution interlock in practice. Stacey, Snowden, and Weick collectively showed the importance of Complexity, Sensemaking, and Feedback Loops for adaptive coherence. McGrath and Eisenhardt further extended this, revealing how Dynamic Capabilities, Heuristics, and Simple Rules operate under conditions of entropy and systemic flux.

In sum, these thinkers contributed fragments that, when repeatedly tested through real-world problems, consistently mapped onto a discrete set of conceptual elements necessary for a general theory: recursion, complex adaptivity, selective pressure, dynamic and entropic informational environments, probabilistic inference, purposeful agency, coherence, and the deliberate orchestration of actions to collapse advantageous futures into realised outcomes. No single framework provided all these elements. Their empirical gaps, systematically revealed through this abductive process, made the final synthesis methodologically necessary.

Importantly, this synthesis is not metaphorical. Each theoretical element drawn into the QCEA paradigm is treated literally and within its ontological boundary. For instance, information theory is used not to evoke metaphorical "signals" but to formalise entropy as measurable uncertainty in inference processes (Shannon, 1948). Quantum principles are not invoked for rhetorical flourish but to model the probabilistic structure of epistemic distributions in uncertain futures, consistent with Bayesian reasoning and quantum cognition literature (Jaynes, 2003; Busemeyer & Bruza, 2012). Complexity science is used not as a general analogy but as an empirical and ontological framework for describing how recursive feedback and adaptive agency produce emergent system-level behaviour (Holland, 1995; Anderson, 1999).

To ensure philosophical coherence across these disciplines, the QCEA framework is governed by a meta-methodological commitment to stratified realism (Bhaskar, 1978; Bunge, 2003). This principle recognises that distinct ontological levels—quantum, epistemic, behavioural, systemic—operate under their own lawful regularities, and can be integrated without collapsing their explanatory functions. Each theory is deployed at the level appropriate to its referent domain. There is no categorical confusion between quantum indeterminacy and macroeconomic uncertainty, nor between emergent complexity and causal determinism. Instead, the framework preserves ontological separation and explanatory complementarity, consistent with established standards for interdisciplinary integration (Mitchell, 2009; Bechtel & Abrahamsen, 2005).

Where some critics might anticipate incommensurability between observer-dependent and realist positions, the QCEA paradigm explicitly resolves this through ontological layering: strategic actors can be epistemically constrained and probabilistically situated (as in Bayesian realism), while remaining embedded in a mind-independent environment shaped by entropic and emergent forces. No single perspective is treated as totalising. Rather, their integration reflects a commitment to non-reductive, abductively validated theory-building, where conceptual robustness emerges through reciprocal refinement across epistemic domains.

This methodological orientation—stratified, abductive, and empirically disciplined—provides the necessary foundation for a non-tautological theory of strategy that is both generalisable and falsifiable. The QCEA paradigm did not emerge from speculative philosophy or rhetorical synthesis. It emerged because strategic problems in real-world environments demanded a coherent, testable explanation for how advantage is cultivated and collapsed under conditions of incomplete, decaying, and contested information.

B. Information as the Ontological Core of Strategy

A pivotal methodological realisation emerged through iterative abductive engagement: strategic action is, at its core, an inferential response to uncertainty, governed by information. Shannon (1948) formalised information as uncertainty reduction; Simon (1956, 1978) revealed the cognitive limits that make such reduction bounded, partial, and consequential. Strategy, thus, is not merely informed by information—it is forged in its incompleteness.

Tetlock and Gardner (2015) empirically confirmed what practice repeatedly demonstrated: strategic forecasts are not defective approximations of a knowable future but necessary distributions over unknowable states. Uncertainty is not a flaw in the method—it is a fact of the epistemic condition.

Clausewitz intuited this. His 'friction' and 'fog' were not metaphors for confusion but structural constraints on inference. Yet, as Freedman (2013) and Strachan (2007) note, Clausewitz stopped short of mechanising how adaptive inference proceeds under such conditions. Von Moltke's Auftragstaktik partially operationalised this insight—delegating judgement to local nodes to reduce epistemic lag—but lacked a formal architecture for coherence under decentralised asymmetry (Howard 1961; van Creveld 1985; Paret 1986).

Sun Tzu likewise grasped the informational substrate—deception, indirectness, and perception-shaping as tools for manipulating belief under opacity (Sawyer 1994; Handel 1991). Yet the Art of War never constructs a systematic epistemology for how recursive belief-shaping reconfigures informational environments. Its aphorisms hint; they do not formalise.

Porter, by contrast, anchored strategy in structure: decoding industry forces to position the firm within a legible landscape (1980, 1996). But this legibility presumes informational stability—an assumption untenable under reflexive, entropic churn (McGrath 2013; Suddaby et al. 2011). His frameworks diagnose structure; they do not adapt to its deformation.

Mintzberg advanced the argument by embracing emergence—strategy as pattern recognition under retrospective signal (1987, 1994). Yet his schools of thought remain classificatory. They elucidate sensemaking, but not its recursive integration into coherent strategic inference (Jarzabkowski 2005; Whittington 2006).

Rumelt brought epistemic clarity to the foreground. Diagnosis, he insists, is the crucible of strategy (2011). But while he distinguishes signal from noise, he leaves the inferential machinery implicit—entrusting coherence to practitioner acumen rather than formal process (Gavetti and Levinthal 2004).

Freedman (2013) synthesises the dilemma: most accounts of strategy "start in the middle"—treating information as context rather than as ontological ground. Gray (1999) and Heuser (2010) echo the critique: the tradition names uncertainty but does not model how knowledge arises within it.

Taken together, these canonical strands converge on a single abductive imperative: strategy is information—fragmented, retrospective, entropic. To theorise strategy is to theorise how agents infer, adapt, and act under conditions of radical epistemic incompleteness. Without a formal, falsifiable framework grounded in entropy, bounded rationality, and probabilistic inference, strategic theory cannot advance beyond heuristics. The absence of such a model is not an

oversight; it is the missing ontology.

This claim is neither metaphorical nor rhetorical. In the QCEA paradigm, information is not treated as an abstract analogy for "insight" or "data," but as a quantifiable ontological substrate—measurable, depletable, and structurally entropic. The epistemological incompleteness faced by strategic agents is not merely cognitive; it reflects a real and irreducible information-theoretic constraint on how strategic landscapes can be apprehended, processed, and acted upon.

To avoid category error, the paradigm distinguishes between syntactic information (as formalised by Shannon) and semantic content (as situated by strategic context). Strategic coherence is not derived from perfect information, but from inferential processes that navigate noise, delay, and ambiguity while conserving coherence under selective pressure. This formulation aligns directly with epistemic realism—the belief that strategy operates over real, external states, but accesses them only via incomplete and decaying signal. Information is thus not the backdrop of strategy; it is its very substance.

The ontological claim, then, is this: strategic action is the epistemically constrained orchestration of intervention in entropic information environments. Coherence emerges not from total knowledge but from the capacity to sustain effective action in the presence of irreducible informational lag. The strategist does not merely interpret the world—they intervene in it to narrow possible futures, collapsing epistemic distributions into realised outcomes.

This methodological position precludes naïve rationalism, circumvents tautology, and replaces metaphor with mechanism. Strategy, under this paradigm, becomes a science of epistemic collapse: the disciplined reduction of probabilistic possibility into actionable reality, via bounded, recursive, and coherence-seeking inference. No other framework to date has modelled this process formally. The QCEA paradigm does so explicitly—and necessarily.

C. Quantum and Probabilistic Foundations

A critical methodological decision was to characterise strategy in fundamentally probabilistic rather than deterministic terms—not as metaphor but as ontological commitment. This commitment emerged abductively from repeated empirical testing and theoretical reflection, but crystallised during a fourth reading of Tetlock and Gardner's Superforecasting (2015). That reading, stress-tested against real-world forecasting failures in strategic advisory contexts, revealed a pivotal insight: the future is not simply unknown; it is indeterminate. For the strategist, reality exists as a distribution of superposed possibilities until deliberate action collapses that uncertainty into realised outcomes.

This insight is formalised—not metaphorically borrowed—via the mathematical apparatus of quantum theory. Schrödinger's (1935) concept of superposition rigorously expresses this condition: that a system exists in multiple potential states simultaneously, represented by a probability amplitude, until an observation collapses the wave function to a single actualisation. The strategist, like the quantum observer, is not a passive diagnostician but an active participant: collapsing entropic futures into realised trajectories by virtue of inferential commitment and action.

Critically, this probabilistic architecture is not merely a philosophical position—it reflects the empirical constraints faced by all inference-making agents. In strategic domains, future market dynamics, adversary intent, policy shifts, or technological adoption pathways cannot be deterministically known in advance. They exist as entangled distributions of possibility. Only by acting—via product launches, investments, or strategic moves—can one collapse those futures into concrete form.

Quantum cognition research (Busemeyer & Bruza, 2012; Khrennikov, 2010; Yukalov & Sornette, 2011) confirms this structure empirically. Human reasoning under uncertainty often violates classical probability theory but conforms to quantum probability formalisms—such as non-commutativity, interference effects, and entangled belief states. These findings are not trivial:

they demonstrate that inference, in both individuals and collectives, often follows probabilistic geometries best captured by Hilbert-space logic rather than Boolean frames.

Bayesian probability, as formalised by Jaynes (2003), reinforces this alignment. Probabilities are not mere frequencies or subjective guesses; they are epistemic structures encoding degrees of belief in possible states of the world, continuously updated as new signals emerge. In this view, strategy is a recursive process of hypothesis, observation, and inferential collapse—governed not by determinism but by uncertainty reduction through bounded rationality.

Heisenberg's (1927) uncertainty principle sharpens this epistemic constraint: it is not possible to simultaneously know a system's position and momentum with complete precision. Translated to strategy, this means that one cannot fully know both the present state of a complex environment and all potential future moves or reactions. This is not cognitive failure but a structural feature of inference under entropic constraint—a limitation that every strategist must navigate.

The Einstein-Podolsky-Rosen paradox (1935) and Bell's (1964) theorem extend this architecture: demonstrating that entanglement—i.e., systemic non-locality and outcome interdependence—is not a violation but a feature of how reality operates under uncertainty. This aligns directly with strategic contexts, where local interventions (pricing changes, policy signals, PR moves) can yield reflexive, non-linear effects far beyond their immediate domain—effects that are not merely correlated but structurally entangled.

Importantly, this framework does not confuse ontological levels. The QCEA paradigm does not claim that firms are quantum objects or that organisations behave like electrons. It claims, rather, that the epistemic structure of strategic inference is homologous to the structure of quantum probability. The mapping is formal, not figurative. The strategic agent navigates a probabilistic ontology, where decisions operate under the same information-theoretic constraints as quantum inference—constraints that are mathematical, not metaphorical.

This position also maintains fidelity to epistemic realism. Quantum mechanics, when framed via ontic structural realism (Ladyman & Ross, 2007), allows for the coexistence of observer-dependent phenomena at the micro-level with stable, mind-independent structures at higher scales. The strategist does not invent futures through sheer belief; they act upon systems with real constraints and structure. What remains indeterminate is not reality itself, but the trajectory that will unfold given bounded knowledge and delayed feedback.

Mingers (1997) warned against superficial borrowing of metaphors from quantum theory into social science. The QCEA paradigm avoids this by adhering to rigorous translation grounded in philosophical stratification: quantum theory informs the structure of epistemic inference; complexity theory frames the meso-level behaviour of adaptive systems; information theory defines the limits of entropy reduction; and strategic theory operationalises action within these constraints.

Thus, within the QCEA paradigm, strategic action is understood as the recursive, probabilistic orchestration of inferential decisions to collapse advantageous superpositions into realised outcomes. The quantum-probabilistic foundation is not rhetorical flourish but methodological necessity. It enables us to formally model what classical strategy theories leave implicit: that advantage arises not from control over outcomes, but from the capacity to act decisively in conditions where the future exists only as a cloud of probabilistic possibility.

D. Complex-Adaptive Systems and Strategic Agency

A further critical methodological advance emerged from the sustained abductive engagement with complexity science, particularly the theory of complex-adaptive systems (CAS). This was not a figurative association but a literal synthesis, grounded in the empirical observation that strategic agents—whether firms, militaries, or states—do not operate in closed systems. They navigate turbulent, interconnected environments characterised by feedback loops, emergent patterns, path dependence, and reflexive dynamics. These properties are not merely descriptive of

context; they are intrinsic to how strategic action functions and must therefore be integrated into any general theory.

Repeated empirical testing in advisory, executive, and organisational contexts consistently revealed that traditional linear planning logics—based on static assumptions, optimisation, and equilibrium-seeking—failed to account for the adaptive, evolving nature of real-world strategy. Rigid plans fractured under conditions of volatility, while systems designed to learn, sense, and adjust recursively—i.e., complex-adaptive systems—were able to maintain coherence and pursue advantage.

John Holland's (1995) foundational work on *Hidden Order* provided the conceptual anchor: CAS are systems composed of interacting, semi-autonomous agents whose local rules and micro-interactions generate emergent macro-level behaviours. These systems cannot be fully understood via top-down control or reductionist decomposition. In the strategic context, this translates directly: firms and institutions generate coherence not through centralised optimisation, but through decentralised adaptation, modular rule-sets, and feedback-rich learning.

Holland's "dancing landscape" metaphor—rooted in Kauffman's (1993) NK models—articulates how fitness landscapes are not fixed but deformed by agents' own movements. Strategic advantage is not located at a stable peak but must be pursued across a topography that shifts as others adapt. This metaphor, though evocative, is grounded in a literal mathematical structure describing rugged, high-dimensional search spaces. In QCEA, this becomes formalised as the recursive re-mapping of advantage under entropic pressure and agentic deformation.

Robert Axelrod and Michael Cohen's *Harnessing Complexity* (1999) advanced this framework into the strategic domain. They demonstrated that adaptive coherence—i.e., system-level order without centralised control—could be cultivated via simple local rules, modular architectures, and feedback loops. This was corroborated repeatedly in practice: organisations that embedded such structures outperformed those that relied on detailed ex ante planning under conditions of environmental flux. This insight forms the cultivation pillar of the QCEA triad (Formulation, Cultivation, Execution): coherence is not imposed; it is recursively nurtured through structure, signal, and local inference.

Waldrop's (1992) chronicle of the Santa Fe Institute's interdisciplinary breakthroughs revealed how this perspective reframed economics, biology, and organisational theory around the same core insight: that systems evolve through recursive adaptation, and that strategic success lies in managing this adaptivity rather than resisting it. In QCEA, this becomes operational: strategy is not the plan—it is the architecture that enables recursive coherence through decentralised inference.

This perspective is strengthened by the work of Herbert Simon (1956, 1978), who demonstrated that bounded rationality necessitates satisficing under uncertainty. Agents cannot optimise across the total landscape; they must adapt heuristically. Strategy, then, becomes the design of heuristics and the feedback structures that allow those heuristics to evolve. March's (1991) exploration of the exploration–exploitation trade-off further supports this: strategic systems must balance novelty search with stability maintenance—another inherently CAS dynamic.

Tsoukas and Chia's (2002) argument that organisations are not static entities but processes of continual becoming reinforces this orientation. They posit that strategy is not a discrete output but an unfolding process of interpretive sensemaking and emergent alignment—what QCEA captures as recursive inference under entropic constraint.

Brian Arthur's (1994) model of increasing returns and path dependence provides a further empirical and theoretical foundation. He shows that small differences in local choices can amplify into systemic lock-ins, bifurcations, or phase transitions. Strategic systems, in this view, must manage not only their current position but the historical trajectories and reinforcing feedbacks that shape their option space. The QCEA framework incorporates this as a structural commitment: strategic action is always recursive, entangled with its own history, and subject to emergent non-

linearity.

Brown and Eisenhardt (1997) and Anderson (1999) extend these insights into the organisational domain, demonstrating empirically that successful firms operating in high-entropy environments employ adaptive architectures: modular teams, semi-structured routines, rapid feedback cycles. These structures allow strategic coherence to emerge and persist even when the macro-environment is in flux. Their findings validate the QCEA proposition that cultivation—rather than top-down imposition—is the only viable pathway to coherence under entropy.

Crucially, the QCEA framework does not treat complexity as an aesthetic description or a cautionary tale. It operationalises complexity theory to build a falsifiable model of strategic coherence in non-equilibrium environments. It treats strategic agents as literally complex-adaptive systems and builds from this a theory of action under uncertainty, in which feedback, path-dependence, inference, and emergent coordination are not exceptions to planning, but its replacement.

Thus, QCEA establishes that strategy is not a linear plan toward a fixed objective. It is a recursive orchestration of agent-based inference, adaptation, and coherence-maintenance in a deforming, entropic landscape. The strategist does not impose order from above, nor merely respond to complexity—they cultivate coherence through architecture, signal design, and recursive action. This reframing, grounded in the literal structures of complex systems theory, enables strategy to be understood as the adaptive navigation of uncertainty—not despite complexity, but through it.

E. Entropy, Information, and Probabilistic Forecasting

A decisive methodological realisation in the QCEA synthesis was the need to ground strategic theory in the literal mechanics of entropy, information, and inference—not as metaphor, but as ontological and epistemological primitives. Strategy, as revealed through repeated empirical testing and abductive synthesis, is not merely influenced by information but constituted by its availability, degradation, and transformation. Under conditions of entropic decay and epistemic incompleteness, strategic coherence cannot be imposed—it must be continuously reconstructed through inferential processes. This insight required direct engagement with the mathematical and physical properties of information itself.

Claude Shannon's (1948) foundational work defines information quantitatively as the reduction of uncertainty in a probabilistic system. Shannon entropy captures the expected value of information content—directly correlating with unpredictability. Crucially, this is not a metaphor: it is a mathematical structure that formalises how knowledge acquisition operates under uncertainty. In strategic contexts, Shannon's formulation enables us to model belief revision, signal reliability, and inferential precision, all of which are essential to strategic action.

Landauer's (1961) principle further strengthens the ontological status of information by showing that erasure incurs a real thermodynamic cost: information is not an abstract substance but a physical constraint. Bennett (1982) and others extend this into the thermodynamics of computation, establishing that information processing—perception, memory, action—necessarily draws upon and dissipates energy. This establishes a non-negotiable entropic budget for strategic activity: all knowledge is partial, all inference degrades, and all coherence must be actively maintained against natural dissipation.

In QCEA, this manifests as the core premise of strategic entropy: coherence is not free, and strategy must account for both the acquisition cost and decay rate of information. The strategist is not optimising within a stable structure but navigating a dynamic landscape of shifting signals, retrospective fragments, and fading relevance. The goal is not certainty, but adaptive coherence under probabilistic constraints.

Boisot's (1998) information-space theory articulates how firms must manage codification, abstraction, and diffusion to shape the informational terrain in which they compete. Snowden and

Boone (2007) build on this to argue that sensemaking architectures must account for fragmentary, ambiguous signals. Their Cynefin framework highlights that decision-making under complexity must be differentiated by domain: simple, complicated, complex, or chaotic. In QCEA, these typologies are unified under a single principle—entropy governs the threshold between these domains, and coherence must be cultivated accordingly.

Bayesian inference provides the mathematical foundation for updating beliefs under such uncertainty. Jaynes (1957, 2003) rigorously formalises this, showing that probability distributions are not mere conveniences but epistemic commitments over plausible states. In the QCEA framework, strategy is understood as a continuous process of belief revision: each action updates the distribution, collapses some possibilities, and opens new ones. The strategist operates not with static knowledge but with evolving probability amplitudes—each shaped by past inference and future expectations.

Tetlock's (2005) longitudinal study of expert political judgment confirms this empirically: single-point forecasts fail systematically, while probabilistic forecasting—updating beliefs with calibrated distributions—correlates with superior strategic performance. Mellers et al. (2000, 2014) extend this, demonstrating that training in probabilistic reasoning enhances forecasting accuracy. Crucially, these findings do not merely endorse a better technique—they reveal the ontological condition of strategic foresight: it is never deterministic; it is always probabilistic and entropic.

Superforecasting (Tetlock & Gardner, 2015) was particularly decisive in the QCEA paradigm's development. A fourth rereading of the text, conducted during a period of empirical failure in strategic forecasting practice, catalysed the abductive insight that forecasting is not defective prediction but strategic inference over superpositions. This resonates with the quantum formalism articulated earlier: the future exists not as a path to be predicted, but as a distribution to be shaped. Strategic action collapses some of these possibilities into realised outcomes—irreversibly, and at an epistemic cost.

The integration of thermodynamic insight here is not symbolic—it is literal. Entropy imposes constraints on how information can be stored, transmitted, and acted upon. Decision latency, misalignment, and interpretive error are not simply managerial challenges; they are thermodynamic realities that degrade coherence. In QCEA, entropy is not noise—it is the governing condition of all strategy. And because entropy accumulates over time, the strategist must not only decide what to do, but when to decide—timing becomes an entropic optimisation.

Floridi (2011) underscores this point by distinguishing between levels of abstraction and modes of epistemic access. Strategy, in this sense, is an inferential architecture for interacting with informationally open systems. It must structure perception, belief, and action across entropic gradients. Classical planning fails not because it is analytically weak, but because it presumes informational stability. In an entropic system, structure must be re-imposed continuously, not assumed.

Taken together, these theoretical and empirical insights compel a new methodological position: strategy is the probabilistic, inferential navigation of entropic information structures. The strategist is not a planner executing a path, but an epistemic agent orchestrating recursive inference under degradation. This is not a metaphor—it is the literal condition of strategic reality. Coherence, in QCEA, is the emergent consequence of recursive belief revision, signal amplification, and timely action under entropic pressure.

Thus, strategy cannot be adequately theorised without engaging the physics of information and the mathematics of entropy. These domains provide not only the constraints but also the structure for a general theory. In QCEA, this becomes operational: strategy is a recursive process of collapsing epistemic uncertainty into realised advantage—guided by information, governed by entropy, and enacted through probabilistic inference.

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F. Evolutionary Epistemology: Validation of Strategic Ontology

To establish strategy as a legitimate ontological category—not merely a managerial artefact—the QCEA paradigm engages evolutionary epistemology not as an analogy, but as a methodological foundation. If strategy claims universality and explanatory power across domains, then it must also be accountable to the most general theory of knowledge acquisition under adaptive pressure: evolution. This section grounds QCEA in that tradition, showing that strategic coherence, adaptive inference, and recursive decision-making emerge precisely because they confer survival advantage under conditions of entropic uncertainty.

Donald Campbell's (1974) foundational articulation of evolutionary epistemology defines knowledge as the emergent product of variation, selection, and retention—a recursive cycle through which systems refine fit to uncertain, changing environments. This is not a metaphorical claim, but a methodological commitment: knowledge does not arise from rational deduction alone, but through iterative feedback, probabilistic testing, and selective persistence. Strategic knowledge operates under the same logic. The strategist generates hypotheses (moves), tests them against reality (feedback), and retains or revises them based on fitness (advantage). QCEA simply formalises this recursive loop using the grammar of entropy and inference.

Plotkin (1994) deepens this argument with his concept of the Darwin Machine: cognitive systems generate internal simulations, evaluate them probabilistically, and privilege those that offer improved predictive control. This recursive architecture precisely mirrors QCEA's conception of strategic formulation, cultivation, and execution. Strategy is not a linear plan—it is a Bayesian engine for updating beliefs under uncertainty, in pursuit of coherence amid flux.

Popper's (1972) falsificationism—often misunderstood as a critique of induction—is in fact a blueprint for adaptive cognition. His model of conjecture and refutation assumes that inferential systems survive only if their beliefs remain open to disconfirmation and revision. The QCEA framework adheres to this principle not rhetorically, but structurally. Strategic systems in QCEA are defined by their recursive capacity to collapse superposed futures into realised outcomes, based on provisional, probabilistic beliefs. These beliefs must be falsifiable—empirically and conceptually—if the system is to remain adaptive.

This methodological insight reframes strategy itself as an evolved cognitive adaptation. Dawkins (1976) makes this plain: systems that process information more effectively persist. In this view, strategy is not an invention—it is a selection-tested response to entropic environments. Strategic agents endure not because they optimise, but because they satisfice under bounded rationality, update recursively, and preserve coherence across probabilistic distributions. Strategy exists because it works—not always, but often enough to be retained and refined.

Dennett's (1995) concept of "design space" further clarifies the terrain. He argues that intelligent systems evolve not by following preordained blueprints but by exploring inferential architectures that enable agency under uncertainty. Strategic capacity—defined in QCEA as the recursive ability to impose coherence and collapse entropy—is one such architecture. It survives because it transforms fragmentary information into navigable action.

Gould (2002) reinforces this position by demonstrating how emergent complexity—co-evolution, path dependence, adaptive modularity—shapes not only morphology but cognition. Strategy, in this context, is not the rational apex of human design; it is the adaptive response to environments where signals degrade, futures fork, and coherence must be earned rather than assumed.

This view is empirically substantiated through cross-disciplinary synthesis. Tooby and Cosmides (1992) argue that key cognitive traits—such as pattern recognition, causal inference, and social coordination—evolved precisely to manage uncertainty in complex, informationally opaque environments. These are not incidental capabilities; they are strategic primitives. In QCEA, they are formalised as recursive inference mechanisms constrained by entropy.

Henrich (2015) extends this claim through cultural evolution. He shows that strategic cognition

scales through transmission: groups accumulate and encode heuristics, enabling coherence across distributed agents. Strategy becomes not merely an individual trait but a population-level adaptation. QCEA incorporates this insight by treating coherence as a cultivated property—not a given, but a systemic achievement.

Tomasello (1999) offers final validation. His work on the cultural origins of human cognition reveals that shared intentionality—recursive coordination across agents—is not a by-product of intelligence but a precondition for strategic action. In QCEA, this is mirrored in the cultivation layer: coherence is scaffolded through shared models, trust architectures, and signal loops that enable distributed inference.

Collectively, these perspectives make a singular methodological claim: if strategy is to be taken seriously as a general theory, it must be accountable to evolutionary constraints. QCEA meets that standard. Its core components—bounded inference, recursive coherence, adaptive collapse—are not managerial preferences; they are survival traits. Each element earns its place not through conceptual elegance but through evolutionary testability.

This reframing resolves a critical philosophical challenge. It avoids the charge of arbitrariness—of layering concepts from physics, complexity, or information science without justification—by demanding that each conceptual element withstand the evolutionary criterion: does this capability enable adaptive action under entropy? If not, it is discarded. If so, it is retained, not as metaphor, but as method.

Thus, QCEA does not merely borrow from evolutionary epistemology—it submits to it. Its architecture is abductively derived, empirically stress-tested, and methodologically falsifiable. Its claims are not shielded by metaphorical abstraction but exposed to selection pressure: does the framework increase an agent's capacity to generate, retain, and act upon knowledge under uncertainty?

That it does—and that no prior framework does so fully—is not a rhetorical flourish. It is the conclusion of a decade-long abductive refinement process grounded in evolutionary necessity. Strategic systems persist not because they predict the future, but because they survive its uncertainty. QCEA is the first framework to formalise that reality without resort to metaphor, managerial fashion, or reductionism. It earns its ontological status by passing the only test that matters: adaptive coherence under entropic pressure.

G. Logical and Syllogistic Validation: Ensuring Theoretical Robustness

If strategy is to mature into a scientific discipline rather than remain an assemblage of heuristics, it must meet the foundational criterion of theoretical legitimacy: falsifiability. A general theory of strategy cannot rely on persuasive rhetoric, metaphorical borrowing, or ad hoc classifications. It must expose its internal claims to contradiction, stress-test their coherence, and establish clear conditions under which those claims would fail. The QCEA paradigm accepts this burden explicitly and was constructed to meet it.

Drawing from Popper's (1959) foundational principle, a theory is scientific not because it is true, but because it can be shown to be false. Conjectures that make no risk of disconfirmation are mere dogma. Within QCEA, every core proposition—strategic superposition, recursive coherence, entropic constraints, adaptive collapse—is structured such that it can, in principle, be refuted through observation, simulation, or comparative failure. The framework does not rely on post-hoc storytelling. It is a diagnostic and predictive system with testable claims.

But Popper alone is insufficient in complex domains. As Lakatos (1976) argues, mature theories are not judged by isolated falsifications but by whether they demonstrate a 'progressive problemshift'—whether they explain more than their predecessors and generate novel, empirically testable predictions. This standard is particularly vital in strategy, a domain long criticised for its reliance on retrospective narrative. Freedman (2013) observed that most strategic frameworks "start in the middle," describing actions without defining the conditions that make them strategic.

Rumelt (2011) similarly lamented that strategy had become a "catch-all term," bereft of boundary-defining logic. QCEA was explicitly developed to meet this gap. It defines strategy not as behaviour, but as a recursive mechanism for collapsing uncertainty into advantage under entropic constraints—a claim that can be tested, disproven, or refined.

To meet this standard, the QCEA paradigm was subjected to continuous logical analysis and syllogistic validation throughout its development. Each conceptual component—be it superposition, adaptive inference, coherence as a systemic attractor, or entropy as the underlying constraint—was articulated as a testable proposition, embedded in a broader inferential grammar. These propositions were interrogated for internal contradiction, alignment with canonical literature, and empirical robustness across cases.

For example, QCEA predicts that strategic systems which fail to maintain recursive feedback between belief updating, environmental scanning, and action will progressively lose coherence and suffer performance decay in complex environments. This is not a theoretical flourish. It is a falsifiable claim, validated against repeated empirical observation. Firms that locked into rigid plans without adaptive feedback—Kodak, Nokia, Blockbuster—collapsed not from lack of vision, but from loss of inferential coherence under changing conditions. Conversely, systems that embedded recursive inference—Amazon, Tencent, adaptive military units—demonstrated strategic resilience through coherence reconstruction.

Further, the paradigm predicts that under conditions of high entropic flux (e.g., technological disruption, geopolitical bifurcation), deterministic strategies premised on static models will systematically underperform compared to those based on probabilistic inference and adaptive cultivation. This claim can be evaluated through simulation, comparative forecasting, and longitudinal performance data. It is a live proposition, not a retrospective generalisation.

Suddaby (2014) cautions that many management theories drift into tautology: they describe what happens, not what must happen, and then restate it as law. QCEA avoids this by explicitly rejecting retrospective sufficiency as a validation criterion. It insists that a theory of strategy must generate prospective tests, grounded in epistemic logic, adaptive necessity, and entropic realism.

The logical structure of QCEA is syllogistic in its foundation and recursive in its method. It begins from the premise that:

- Strategic agents act under uncertainty.
- Uncertainty arises from entropic information environments.
- Advantage accrues to agents who collapse uncertainty into coherent outcomes through adaptive inference.

From this, QCEA derives a series of operational corollaries:

- Strategic action is probabilistic, not deterministic.
- Coherence is not given; it must be cultivated and maintained.
- Systems lacking recursive feedback degrade over time.
- Entropy is not a metaphor—it is the constraint condition that defines the cost and possibility of inference.
- Superpositions describe the real condition of futures—not because of physics, but because of epistemology.

Each corollary is falsifiable, not merely by counterexample but by theoretical contradiction or failure in applied use. The framework was repeatedly tested across strategic domains—corporate, military, public policy, and investment environments—each time exposing weak assumptions, refining operational language, and rejecting any construct that could not withstand disconfirming evidence or internal tension.

Importantly, QCEA's logical foundation does not rest on disciplinary insulation. It does not assume that concepts like coherence, entropy, or inference retain their integrity when casually translated. Rather, it constructs its grammar through principled integration. Coherence is not a

buzzword; it is an emergent systemic attractor grounded in complex adaptive dynamics (Holland, 1995). Entropy is not a metaphor for disorder; it is a formal measure of uncertainty derived from information theory (Shannon, 1948) and thermodynamics (Landauer, 1961). Superposition is not physics theatre; it is the ontological status of uncollapsed futures as experienced by bounded agents acting under incomplete information (Jaynes, 2003; Tetlock & Gardner, 2015).

This method was not adopted to impress, but to survive scrutiny. It is the only way to avoid the category errors that have historically undermined interdisciplinary theorising. By committing to logical clarity, empirical testability, and rigorous inference, QCEA avoids the twin traps of metaphorical inflation and theoretical vagueness.

It is worth stating plainly: QCEA does not describe what strategy *looks like*; it describes what strategy *is*—and how that can be tested. It treats strategic agents not as rational optimisers or heuristic gamblers, but as inferential engines navigating entropic landscapes. Every element of the paradigm—coherence, collapse, adaptation, recursion—is open to challenge. This is its strength, not its weakness. It welcomes contradiction, because only through contradiction can theory evolve.

In this, QCEA does not seek immunity from critique. It seeks exposure. It offers a structure precise enough to be disproven, yet robust enough to survive. That is what separates a theory from a slogan, and a paradigm from a passing framework.

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H. Limits and Directions for Future Refinement

The QCEA paradigm as presented is neither final nor exhaustive. It offers a falsifiable, interdisciplinary theory of strategy, but its advancement depends on further empirical, formal, and conceptual development. Three areas, in particular, warrant deeper investigation:

Teleological Explanation: While QCEA incorporates purpose as an emergent outcome of adaptive coherence, the interaction between purposeful agency and probabilistic structure remains under-theorised. Future work must engage more deeply with normative epistemology to clarify how intent, values, and goal-setting interface with recursive inference.

Nested Agency and Multi-Level Systems: Strategic action rarely resides in a single actor. Firms exist within alliances; states within institutions; teams within organisations. The boundaries of strategic agency in multi-agent systems—particularly under asymmetric information and distributed authority—require formal modelling. This demands further development of coherence principles under nested recursion and cross-scale feedback.

Simulation-Based Validation: While the QCEA paradigm is constructed to be logically consistent and empirically grounded, its predictive implications have yet to be formalised through simulation. Agent-based modelling and entropic Monte Carlo methods offer promising avenues for testing how strategic coherence emerges, decays, or bifurcates under varying informational and environmental conditions.

These are not disqualifying gaps but invitations for iterative refinement. Each reflects the paradigm's core principle: strategy must remain adaptive, recursive, and open to correction. The QCEA framework is not an endpoint—it is a structured platform for ongoing inquiry.

I. Iterative Qualitative Synthesis and Abductive Reasoning

QCEA does not offer a collage of metaphors, nor a pluralist inventory of perspectives. It offers a unified, falsifiable, and ontologically coherent framework for understanding strategy as it is practiced in real-world systems—biological, organisational, geopolitical, or artificial.

Each foundational domain plays a defined, irreducible role:

- **Quantum theory** articulates the structure of epistemic uncertainty—the probabilistic superposition of futures prior to action.
- **Complexity theory explains** how coherence and adaptive behaviour emerge from decentralised interaction and feedback.
- **Information theory** formalises the decay of signal, the cost of inference, and the limits of strategic foresight under entropic conditions.
- **Epistemic realism** secures the ontological status of strategic environments and actions as mind-independent, even as our access to them is bounded and probabilistic.

Together, these elements constitute more than a theory of choice or a model of planning. They define strategy as a recursive collapse of uncertainty into coherent outcomes—a disciplined, adaptive process of shaping futures that cannot be known in advance, yet must be navigated all the same.

QCEA is not a metaphor. It is not a metaphor for war, nature, chess, or quantum mechanics. It is a literal, testable framework constructed for systems that must act amid uncertainty, infer from noise, and adapt under pressure—systems that must impose coherence upon an entropic world without ever fully controlling it.

This is the condition under which strategy arises. And this, precisely, is what QCEA is built to explain.

A. Information as Substrate: Boundedness, Entropy, and the Arrow of Time

Strategy, in its universal sense, begins not with abstract plans but with information — as physical substrate. Shannon (1948) defined information as the measurable reduction of uncertainty, directly extending Boltzmann's statistical mechanics: entropy quantifies the number of microstates consistent with a macrostate. Brillouin (1956) and Jaynes (1957) formalised this equivalence, showing that Shannon entropy and Gibbs–Boltzmann entropy are mathematically identical when a bounded system processes probabilistic states.

Landauer (1961) established the thermodynamic anchor: information is physical. Every act of storing, erasing, or processing data exacts an irreducible energetic cost. Bennett (1982) extended this principle: no computation, however elegant, escapes thermodynamic penalty. MacKay (2003) and Schneider (2010) confirm that even biological systems — DNA transcription, neural signalling, cellular regulation — obey this cost structure. Bekenstein (1981) and Hawking (1974) push to the cosmological limit: the Bekenstein bound formalises how much entropy — and thus how much information — any finite region can hold. No system, biological or artificial, can store or compute the totality of its environment.

The arrow of time gives this substrate its direction. Penrose (1989) and Carroll (2010) show that the universe's low-entropy beginning fixes the temporal asymmetry on which causal inference depends: the past constrains the present; the present encodes probabilistic futures. Earman and Norton (1998) affirm that the second law of thermodynamics holds across all bounded, energy-dependent structures that resist equilibrium. Time, under this constraint, is not mere chronology. It is the entropic gradient that makes learning both necessary and partial.

This is not abstract physics but the scaffolding of adaptive behaviour. Smith and Szathmáry (1995) and Martin (1999) demonstrate that each major evolutionary leap — from replicators to multicellular organisms — is a local solution to this universal drift: systems persist only by capturing and refining retrospective signals to navigate uncertainty. Campbell (1974) and Plotkin (1994) extend this to epistemology: knowledge is not a fixed archive but an evolved apparatus for managing bounded, degrading signals under directional time.

Strategy exists because information is physical, bounded, and decaying. No system can process or retain total knowledge of its environment. Strategic action arises as a response to this universal constraint: the need to transform fragmentary and irreversible signals into temporally aligned,

resource-bound interventions.

Any system that endures — firm, species, algorithm, or state — must continuously extract, encode, and revise partial information under irreversible time. Strategic capacity is bounded by its ability to maintain coherence faster than entropy degrades it. The operational implication is precise: where informational decay outpaces inferential adaptation, strategy fails — irrespective of resources or intent.

Consider BlackBerry in the mid-2000s. Despite possessing extensive internal data on consumer behaviour and market trends, it failed to decode the decisive signal: that general-purpose touchscreen devices were collapsing the relevance of its core value proposition. The information was available. What failed was the ability to extract, encode, and revise it rapidly enough to maintain coherence as the environment drifted. The collapse was not due to ignorance, but to bounded, delayed, and entropic information processing.

B. Quantum Uncertainty and Probabilistic Futures

Heisenberg's (1927) uncertainty principle sets a fundamental constraint: certain property pairs — such as position and momentum — cannot be simultaneously known with arbitrary precision. This is not a measurement error but a physical boundary: the more one knows of one property, the less one can know of its conjugate. Earman and Norton (1998) confirm this interpretation as ontological, not epistemic: uncertainty is built into the fabric of the universe.

Schrödinger's (1935) formulation of superposition clarifies the decision-relevant implication: until interaction occurs, a system exists in a distribution of possible states. Action collapses this distribution. In strategic terms, this implies that any decision requires commitment before full certainty is available.

Empirical support for this logic extends beyond physics. Bruza et al. (2009) and Wang et al. (2014) show that human cognition exhibits interference effects structurally analogous to quantum systems: beliefs shift probabilistically, not deterministically. Busemeyer and Bruza (2012) and Khrennikov (2010) confirm that decision processes — especially under volatility — reflect overlapping, non-commutative probability amplitudes. Pothos and Busemeyer (2013) and Yearsley (2017) contest the extent, but not the existence, of these effects. The point is not metaphorical. Probabilistic superposition is the formal grammar of belief under uncertainty.

Bell's (1964) theorem, resolving the Einstein-Podolsky-Rosen paradox, affirms a deeper constraint: entangled systems cannot be understood in isolation. The informational state of one element conditions that of another, irrespective of spatial distance. Nielsen and Chuang (2010) demonstrate that this principle governs quantum information theory. In complex systems (Anderson 1999; Snowden 2002), this constraint implies that local decisions — even when made with bounded information — can reshape the entire system's topology in unpredictable ways. Strategic foresight must account for entanglement: decisions are conditioned by, and in turn condition, environments that are partially opaque.

Jaynes (2003) reframes this into epistemology: Bayesian updating is not simply a statistical heuristic. It is the only consistent mechanism by which systems can revise belief under fragmentary evidence. Tetlock (2005) and Mellers et al. (2014) provide empirical support: forecasters who hold and refine distributions of belief outperform those who anchor to fixed priors. Tetlock and Gardner (2015) confirm that superior prediction arises not from confidence, but from disciplined probabilism. Strategic planning, under this view, cannot be a linear forecast. It must be an inferential process, anchored in distributions that update with each signal.

Any agent that acts strategically must collapse probabilistic informational states into committed outcomes without access to full certainty. The testable prediction is that systems which update beliefs probabilistically, and act without waiting for certainty, will outperform those which delay commitment in pursuit of false precision — particularly under volatile or entangled conditions.

Kodak's decline illustrates that possessing information does not ensure foresight. The company's own engineers invented the digital camera, and internal forecasts identified its disruptive potential. Yet strategic commitment was delayed. The firm treated future states as predictable extrapolations from the present, failing to act while the probability cloud narrowed. The collapse was not due to absence of data, but failure to act under conditions of probabilistic uncertainty — where belief must anticipate commitment, not follow confirmation.

Strategy does not operate under certainty. Any agent that holds information does so within a probabilistic envelope. Acting strategically requires collapsing possible futures into irreversible outcomes without access to full information — a process bounded by the limits of quantum logic, not deterministic prediction.

C. Emergence of Adaptive Agents

Holland (1995) defines complex-adaptive systems as those composed of local agents operating under heuristic rules, continually modified by feedback. Mitchell (2009) and Gell-Mann (1994) confirm that coherence in such systems arises not from central control but from distributed, rule-based adaptation. Without this architecture, information cannot be transformed into persistence.

Axelrod (1984) demonstrated that even minimal local rule-adjustment enables cooperative advantage. Kauffman (1993) adds the terrain: agents operate on "NK landscapes," where every strategic move deforms the fitness gradient beneath it. Strategic advantage is not found, but actively remade through interaction.

Thompson (2005), in co-evolutionary ecology, affirms that such adaptive interaction is systemic: no agent adapts in isolation. Local adaptations alter global conditions, forcing further recalibration — a recursive dance of action and response.

Simon's (1956) bounded rationality explains the mechanism: agents do not optimise but satisfice, deploying heuristics under constraint. March's (1991) "exploration vs. exploitation" framework captures the oscillation: strategy must balance rule refinement with strategic novelty to maintain coherence under changing conditions.

This is not rhetorical emergence. Anderson (1999) and Snowden (2002) warn that the term "complexity" can become vacuous without structural constraint. Here, emergence is strictly defined: it arises from the recursive modification of local rules under feedback within bounded informational environments.

Strategy requires more than information. It demands an agent capable of recursively transforming uncertain inputs into adaptive action — a complex-adaptive system that persists by updating internal rules in response to external drift.

Strategic systems that exhibit complex-adaptive architecture — defined by decentralised heuristics, recursive rule adjustment, and feedback integration — will outperform rigid systems in volatile environments. The empirical prediction is that coherence persists only where inference, adaptation, and feedback are structurally embedded.

Blockbuster's strategic failure was not informational but architectural. The firm possessed data on consumer trends and digital transition but lacked the adaptive architecture to revise embedded routines. By contrast, Netflix operated with a recursive model — adapting pricing, delivery, and content in response to shifting signals. Strategic coherence did not emerge from pre-existing plans but from adaptive feedback embedded within the firm's operational structure.

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D. Knowledge and Inferential Reach — Universally Expanded

Information, however physically grounded, is inherently retrospective. It arrives from the past, shaped by prior events and degraded by entropic drift. But strategy cannot operate backwards. It requires inference about future states, based on signals from the past. This asymmetry introduces

a critical distinction: information describes what was; knowledge constructs what could be.

This act is not optional. For a system to persist, it must transform fragmentary past signals into provisional belief structures—models that narrow possible futures and guide coherent action. This minimal function is what defines knowledge: justified belief under uncertainty (Plato, Theaetetus). Gettier (1963) and Hetherington (2001) refine this: the value of belief lies not in invulnerability, but in adaptive utility—it need not be infallible, only improvable.

This capacity predates sentient minds. Monod (1971) shows proteins responding to molecular gradients: a primitive epistemology of conditional action. Maynard Smith (1982) and Koonin (2011) reveal error correction in viral replication—rudimentary inference systems designed to preserve pattern despite drift. Campbell (1974) and Plotkin (1994) generalise this to evolutionary epistemology: variation is noise, selective retention is the filter that constructs predictive belief.

The pattern recurs across biological scales. Insects and fish parse signal from noise through sparse architectures (Dukas 1998); fish schools collapse distributed cues into predator inference (Ward et al. 2008); ant colonies (Franks et al. 2002) pool individual scouts' assessments into emergent belief—coordinated action arising from decentralised drift.

Among primates, inference from past coalitions (De Waal 1982, 2016; Whiten & Byrne 1988) and recursive mind-modelling (Tomasello 1999) scaffold multi-agent knowledge systems. Human cultures extend this further: Henrich (2015), Boyd & Richerson (2005), and Heyes (2018) show that culture itself functions as a distributed inference engine, recursively testing, transmitting, and refining belief systems to stabilise action under uncertainty.

Yet none of this arises without the complex-adaptive substrate. As Gell-Mann (1994) and Mitchell (2009) confirm, feedback, rule revision, and emergent pattern recognition are prerequisites for conditional inference. Knowledge is not archived fact, but recursively calibrated belief—and only complex-adaptive systems possess the architecture to perform this transformation at scale.

The quantum constraint still binds. Because all information is fragmentary and entangled, knowledge cannot deliver certainty—it can only narrow the field of probable futures. Robust belief structures contract the superposition; poor inference leaves systems paralysed by volatility. Thus, knowledge is not epiphenomenal—it is elemental: the structural map by which adaptive systems transmute retrospective signals into forward-oriented coherence.

Information is retrospective. Knowledge is the system's capacity to convert retrospective signals into provisional belief about possible futures. This requires inference: selecting, refining, and aligning data under entropic drift to generate coherence.

Information encodes the past. Yet action targets the future. Therefore, strategy cannot rest on information alone. It requires an inferential mechanism that transforms fragmentary, decaying retrospective signals into a conditional forecast that guides adaptive action. This mechanism is knowledge — not memorised code but recursive inference under constraint. Without this layer, coherence degrades and strategic failure becomes likely, even when data is abundant.

Strategy is not based on perfect foresight, but on the construction of knowledge from imperfect retrospection. It is the recursive orchestration of justified belief that allows adaptive systems to collapse entangled futures into realised advantage. Without epistemic transformation, information remains inert.

Systems that explicitly structure the transformation of retrospective information into actionable belief—through recursive updating, feedback, and distributed calibration—will outperform systems that treat information as static input or treat knowledge as fixed. Strategic efficacy correlates with the speed and precision of inference over memory. Among entities with access to similar information, those with stronger inferential architectures (knowledge systems) will exhibit higher coherence, faster response to drift, and greater long-run persistence under volatility.

IBM and DEC both observed the early signals of the personal computing revolution. However, IBM treated the PC as a potential strategic inflection point and developed the IBM PC through an

independent unit, hedging institutional resistance. DEC dismissed the shift, anchored in its belief that minicomputers would remain dominant. Both firms had access to similar signals; only one transformed that information into coherent belief and action. DEC collapsed. IBM reinvented.

E. Selective Pressure and Coherence

Information, belief, and adaptive inference form a recursive loop—but coherence alone is not enough. Evolution does not reward internal sophistication unless it produces external advantage. Selective pressure is the final filter: the empirical reality that deletes incoherent strategies regardless of their elegance. Darwin (1859) codified this principle: variation without advantage is erased. Mayr (1982) unified the dynamic—drift, mutation, selection—as a singular test of persistence. Futuyma (1998; 2013) confirms the point: selection punishes incoherence, not complexity. Williams (1966) formalised the boundary: only traits that enhance net fitness survive. Hamilton (1964) extended this to alignment: systems propagate coherence across agents, or fragment into failure.

The Red Queen hypothesis (Van Valen, 1973) crystallises the entropic edge: systems must adapt merely to stay level—stasis is erosion in disguise. Campbell (1974) and Plotkin (1994) recast this pressure epistemically: belief systems that fail to refine signal degrade into noise. Dawkins (1976) shows that at the gene level, fidelity is not a convenience—it is selected. Dennett (1995) and Gould (2002) confirm that strategic leaps come not from drifted accumulation but from recursive correction: feedback loops that recalibrate coherence against a changing landscape.

Selective pressure adjudicates not the presence of parts, but the orchestration: the fidelity, timeliness, and coherence with which a system gathers information, refines belief, and coordinates action. In a volatile environment, this orchestration must collapse uncertainty faster than entropy degrades alignment.

Among systems with comparable complexity, the one with faster coherence restoration under informational drift will persist longer under selective pressure, all else equal.

Kodak and Fujifilm both detected the shift toward digital photography in the 1990s. Kodak, despite extensive R&D and early patents, failed to integrate signal parsing, belief revision, and organisational alignment fast enough to adapt. Fujifilm restructured aggressively, developed new capabilities, and diversified based on signal coherence. The result: one collapsed, the other persists. Selective pressure did not test their awareness—it tested the speed and accuracy of their coherence loop.

F. Recursion, Feedback, and the Dancing Landscape

No adaptive agent acts on a static stage. Every move alters the landscape on which the next move depends. Kauffman's (1993) NK landscapes show this mathematically: each decision perturbs the topography of advantage. Levin (1998) and Holling (2001) confirm that ecosystems and organisations alike inhabit rugged, shifting terrains, where inference must continuously re-align with emergent contours.

Van Valen's (1973) Red Queen Hypothesis anchors the same principle biologically: in co-evolving systems, survival demands perpetual recalibration. Predator-prey dynamics, host-parasite interactions, and mutualistic loops (Ridley 1993; Dawkins 1982) exemplify this: drift does not pause—strategic sophistication must co-evolve with the system's inferential capacity.

Holland (1995) operationalised this as the defining trait of complex-adaptive systems: survival depends on feedback loops that revise local rules in light of emergent outcomes. Mitchell (2009) and Gell-Mann (1994) affirm that recursion is not cosmetic—it binds local adjustment into global coherence. Thompson (2005) shows that adaptive agents reshape shared terrain, forcing each to iterate not in isolation, but within a coupled inferential web.

This is not a linear trajectory but a punctuated one. Maynard Smith and Szathmáry (1995) map evolutionary transitions—from replicators to social superorganisms—as discrete leaps in information architecture. Gould and Eldredge (1977) confirm these are not gradual accretions, but sudden increases in structural capacity: leaps in feedback fidelity, memory, and inference. Donald (1991) on symbolic storage, Tomasello (1999) on cooperative mindreading, and Shettleworth (2010) on comparative cognition all show that each expansion of informational reach sharpens strategic recalibration.

Across lineages, the same pattern repeats. Ant colonies aggregate scouts' data into distributed consensus (Franks et al. 2002). Fish schools perform decentralised forecasting from minimal cues (Ward et al. 2008). Primates adjust coalitions in response to dynamic social signals (De Waal 2016; Whiten & Byrne 1988). Henrich (2015) and Heyes (2018) extend this to human cultural evolution: social learning and scaffolding act as recursive amplifiers of collective inference.

Holling's (2001) adaptive cycles show the cost of stasis: without renewal, coherence dissolves. The same logic applies in artificial systems. Holland (1995) anticipated this: genetic algorithms and machine learning embed recursive rule adjustment into their architecture. Mitchell (2019) and Russell & Norvig (2021) confirm this empirically: reinforcement learning, ensemble models, and real-time AI all function by co-evolving inference with shifting data inputs. Floridi (2011) frames this as the informational frontier: in environments saturated with signal and noise, only those systems with robust, recursive inference loops can collapse uncertainty into strategic advantage.

The sophistication of strategy co-evolves with the system's capacity for recursive inference and environmental recalibration. Advantage is not fixed; it drifts, and only systems that update faster than coherence erodes can persist.

The effectiveness of a strategic system under volatility is a function of the fidelity, speed, and adaptability of its recursive feedback loops—not of its initial plan.

Toyota's lean production system exemplifies recursive strategic logic. Unlike Ford's deterministic assembly model, Toyota embedded feedback at every node—continuous recalibration, kaizen loops, and local decision rights. When supply shocks or quality deviations emerged, the system adapted in real time. The strategy wasn't the plan—it was the loop. Over time, this recursive edge allowed Toyota to outlearn and outlast static rivals..

G. Information as Substrate: Boundedness, Entropy, and the Arrow of Time

Consider BlackBerry in the mid-2000s. Despite possessing extensive internal data on consumer behaviour and market trends, it failed to decode the decisive signal: that general-purpose touchscreen devices were collapsing the relevance of its core value proposition. The information was available. What failed was the ability to extract, encode, and revise it rapidly enough to maintain coherence as the environment drifted. The collapse was not due to ignorance, but to bounded, delayed, and entropic information processing.

Strategy exists because information is physical, bounded, and decaying. No system can process or retain total knowledge of its environment. Strategic action arises as a response to this universal constraint: the need to transform fragmentary and irreversible signals into temporally aligned, resource-bound interventions.

H. Coherence as a Collective Amplifier

Coherence is not decorative but elemental: it transforms fragmentary, local inference into system-wide advantage under drift. Maynard Smith and Szathmáry (1995) trace this through biology's major transitions — from genes to chromosomes, unicells to multicellulars, solitary actors to eusocial colonies — showing that evolutionary leaps coincide with improved alignment mechanisms. Persistence arises not from power but from synchronisation: when parts suppress

conflict, exchange signals, and integrate feedback, isolated signals fuse into emergent order.

Uhl-Bien and Marion (2008) transpose this principle into organisations: coherent systems do not eliminate autonomy, but synchronise it. Where agents operate under fragmented rules, drift dissolves direction; where coherence binds them, complexity becomes leverage. Boisot (1998) applies the same logic to informational architecture: coherent structures reduce uncertainty at scale by synchronising interpretive schemas and avoiding semantic conflict. In doing so, they collapse probabilistic futures into realised outcomes.

Axelrod (1984) confirms that even minimal cooperation outperforms defection when reciprocity and alignment persist across repeated games. Dawkins (1976) makes the same case at the gene level: coherence between replicators ensures lineage survival; defectors pay the price of fragmentation. Coherence binds self-interest into emergent stability, functioning as the invisible membrane that transforms divergence into systemic resilience.

Empirical patterns affirm this logic. Slime moulds aggregate disparate cues into directional movement. Ant colonies (Franks et al. 2002) and fish schools (Ward et al. 2008) outperform solitary search by pooling, filtering, and aligning fragmentary signals. These are not metaphors—they are living demonstrations that coherence transforms weak local inference into robust global coordination.

Coherence is the structural amplifier of strategy: it binds fragmentary signals, synchronises feedback, and transforms decentralised agents into a system capable of resisting entropic drift.

The strategic persistence of a system increases with the degree to which its information-processing structures enable distributed, synchronised inference across agents.

Spee & Jarzabkowski (2011) analyse how Toyota maintained strategic coherence through tightly coupled strategy tools, routines, and cultural alignment. They note how practices like continuous improvement (kaizen) and just-in-time (JIT) were embedded across layers of the organisation.

Cusumano (1985) and Liker (2004) show how Toyota's TPS (Toyota Production System) was not a modular technique but a deeply integrated system of practices, beliefs, and feedback loops that reinforced each other—aligning operational, cultural, and strategic dimensions.

Dyer & Nobeoka (2000) provide evidence that Toyota's supplier network exhibited coherence through information sharing norms and routines that allowed local autonomy while preserving systemic advantage.

Echevarria (2007) in "Clausewitz and Contemporary War" emphasises the Prussian model's insistence on Auftragstaktik—mission command—which delegated tactical flexibility while preserving strategic intent through shared understanding. This allowed decentralised decisions to remain coherent with strategic aims.

Van Creveld (1985) and Naveh (2007) both document how the Israeli Defence Forces, through the development of Systemic Operational Design (SOD), attempted to restore strategic coherence in complex theatres by embedding cognitive alignment across command structures.

Builder (1989) shows how the U.S. Air Force, in contrast, often suffered from "strategic incoherence" when organisational identity overrode strategic logic, privileging technical preference over adaptive alignment.

Luttwak (2001) distinguishes between tactical excellence and strategic coherence, noting that many failed campaigns (e.g., Vietnam) reflected a breakdown in the alignment of political aims, military operations, and organisational feedback loops.

I. Design, Purposeful Agency, and Non-Survival Goals

Design and foresight mark the crossing from reactive adaptation to purposeful orchestration.

Where early strategy was a recursive response to entropic drift, the advent of cognitive sophistication allowed agents to anticipate, shape, and stabilise desired futures. Dennett (1995) crystallises this: design is not a divine blueprint but the emergent capacity of adaptive systems to simulate conditions yet unrealised — to sculpt landscapes before drift imposes its verdict.

The biological record traces this progression. Protocells sensed chemical gradients but could not plan. Eusocial colonies built resilient architectures — termite mounds, biofilms — where coherence of form followed selective function. Yet foresight remained embedded in embodied rules, blind to abstract ends. With the rise of brains, memory, and symbolic reasoning, strategy shed its purely survival-bound telos. Tomasello (1999) shows that joint intentionality allows primate groups to pursue aims detached from immediate survival — alliances, status, ritualised conflict — each an abstract construct rooted in inference. Henrich (2015) extends this: cultural institutions, stories, and shared norms amplify the collective imagination, enabling designs that persist beyond individual life.

The emergence of non-survival strategic goals is no trivial embellishment but a fundamental shift. Aesthetic appreciation, exploration, knowledge for its own sake — these pursuits become artefacts of surplus cognitive capacity. They reveal that once agents wield foresight, they cease to be mere respondents to drift. They become architects of futures not strictly dictated by immediate selection.

Design thus becomes an artefact of strategy's recursive loop. To design is to hold a model of the possible, to test it inferentially against entropic decay, and to intervene to stabilise preferred superpositions. This phenomenon is not confined to biological lineages: each innovation in communication — storytelling, writing, data — extends the horizon of shared foresight. Organisations, companies, and institutions are not spontaneous aggregations but designed collectives — entities engineered to pursue objectives that abstract beyond the individual's survival calculus.

Strategy has evolved from supporting survival persistence to the pursuit of abstract non-survival goals, but survival of the strategic organism remains a necessary condition for the successful pursuit of these non-survival goals.

Systems that fail to maintain recursive coherence under drift—regardless of their pursuit of higher-order, non-survival aims—will experience collapse of both survival capacity and strategic purpose. The persistence of the strategic agent remains the necessary substrate for realising abstract objectives.

The downfall of WeWork exemplifies this principle. Though the organisation pursued a visionary, purpose-driven agenda—"elevating the world's consciousness"—it failed to maintain coherence between resource allocation, strategic narrative, and operational survival. The abstract strategic goals could not be realised because the system lost the ability to persist. In contrast, Patagonia successfully pursues environmental and social goals while maintaining financial and operational resilience—demonstrating that abstract purpose can be sustained only when the organism itself remains strategically viable (Bocken et al., 2014; Moon, 2020).

J. Communication, Culture, and Collective Strategic Groups

Where individual strategy is bounded by the limits of a single inferential loop, communication and culture emerge as the structural means by which foresight scales. Language, narrative, and ritual are not social ornament but alignment technologies: they bind fragmentary signals into synchronised collective inference robust enough to resist drift.

Boyd and Richerson (1985) formalised this principle through their model of cultural evolution as an adaptive system. Social learning functions as a selective amplifier: by retaining advantageous heuristics and pruning maladaptive noise, culture enables strategic refinement beyond the limits of isolated trial-and-error. This core insight is extended by Henrich and McElreath (2003), who demonstrate that under volatility, social learning systematically outperforms individual learning

— especially when cultures evolve mechanisms to filter free riders and transmission errors. Henrich (2015) expands this into a civilisational thesis: societies with dense, high-fidelity transmission outperform those with shallow, insular loops. Yet the field does not converge into consensus. Richerson and Boyd (2005) acknowledge that cultural drift, norm conflict, and informational corruption threaten coherence. For strategy, this tension is critical: culture must be robust enough to filter noise, yet flexible enough to recalibrate under emergent conditions.

Axelrod (1984) remains foundational in demonstrating how primitive reciprocity can scale into durable cooperation. His iterated tournaments revealed that simple rules, reinforced by memory and signalling, outperform pure defection when interactions recur. Later work complicates but strengthens the point. Nowak and Sigmund (1998) show that indirect reciprocity — through reputation systems and gossip — allows groups to stabilise cooperation even under imperfect monitoring. Bowles and Gintis (2011), building on critiques such as Binmore (1994), argue that stable cooperation requires not only shared norms but credible punishment and nested institutional structures. The relevance to strategic coherence is explicit: trust, signalling, and norm enforcement act as cultural modules that bind distributed inference into robust collective patterns.

Tomasello (1999) situates this insight at the cognitive threshold. Joint intentionality — the capacity to share mental models — enables agents to coordinate on futures that no single mind could simulate alone. This phenomenon is not uniquely human. Whiten and Byrne (1988), De Waal (2016), and others demonstrate that social learning and tactical deception extend across species: chimpanzees, corvids, elephants, and dolphins all exhibit forms of collective inference shaped by shared cues. Heyes (2018) sharpens the theoretical claim: these shared inferential capacities are not innate, but culturally scaffolded — "cognitive gadgets" tuned by social transmission. While some argue this diffuses agency too widely, the comparative record affirms that wherever shared inference architectures emerge, strategic coherence scales.

Empirical support extends across the animal kingdom. Dukas (1998) and Shettleworth (2010) document how fish schools, insect colonies, and bird flocks aggregate local cues into distributed decision-making systems. Bonabeau, Dorigo, and Theraulaz (1999) codify this as swarm intelligence: decentralised agents can infer and act upon dynamic landscapes more adaptively than monolithic planners — especially where volatility penalises rigidity.

In organisational theory, this logic is mirrored. March (1991) formalised the tension between exploration and exploitation: strategic coherence must preserve accumulated knowledge while continuously probing the unknown. This insight seeded a vast literature — from Levinthal and March's (1993) models of organisational learning to Snowden's (2002) Cynefin framework — highlighting that communication structures mediate the balance between rigidity and adaptability. Argyris and Schön (1978) reinforce that learning organisations depend on real-time feedback loops to expose hidden errors and recalibrate inferential structures.

Boisot (1998) completes the picture with a structural lens: cultures do not merely transmit content but shape the very codebook by which signals are interpreted. This is echoed in the literature on sensemaking. Weick (1995) argues that coherence is not passive consensus but the active synchronisation of divergent local interpretations. Where this alignment fails, strategic foresight fragments; where it holds, distributed inference compounds across the system.

Communication and culture scale strategy by synchronising distributed inferential loops into collective foresight. They do not merely transmit behaviour — they filter, scaffold, and align belief structures robust enough to resist drift across a coordinated group. Strategic coherence emerges not from consensus but from recursive calibration of divergent perspectives into shared direction.

Systems that embed mechanisms for high-fidelity signal transmission, error correction, and inferential alignment across agents will outperform those that rely on isolated inference, especially under conditions of environmental volatility or informational ambiguity.

The honeybee colony exemplifies decentralised strategic coherence through communication and cultural transmission. Individual scout bees perform waggle dances to encode spatial vectors to resource sites. The frequency, angle, and duration of these dances serve as probabilistic signals

— not commands — that guide other foragers' exploration (Seeley, 2010). This collective inference process — aggregating noisy, local data into emergent consensus — enables the hive to adapt fluidly to changing landscapes without central control. The system is robust to loss, adaptive to perturbation, and capable of exploratory recalibration when conditions shift. As Seeley and Visscher (2004) show, this alignment of autonomous agents through recursive signalling produces decisions that often exceed the performance of centralised optimisation — a biological instantiation of distributed strategic foresight under entropic drift.

W. L. Gore's decentralised innovation model exemplifies strategic coherence through culture. Eschewing formal hierarchy, the firm employs a "lattice" communication system where self-organising teams align around emergent opportunity, not imposed plans. Heckscher and Adler (2006) document how the firm's norms of transparency, trust, and reciprocal accountability produce a collaborative community that adapts continuously to external drift. Strategic foresight arises not from centralised foresight but from distributed inference embedded in cultural scaffolding.

The German military doctrine of Auftragstaktik institutionalised decentralised autonomy within a shared cultural framework. Officers were trained to understand strategic intent and act independently within volatile battle conditions. As Van Creveld (1985) and Naveh (2009) show, this model relied on trust, doctrinal alignment, and implicit coordination—enabling high-speed adaptation across fragmented units. Strategic coherence emerged from cultural training and communication rituals, not central control.

Whales and dolphins use communication, imitation, and social transmission to propagate distinct foraging strategies, dialects, and cooperative behaviours. Whitehead et al. (2004) show that killer whales maintain pod-specific hunting traditions passed down through generations. These strategic repertoires evolve through culture, not genetics—demonstrating that collective inference and adaptation can emerge from socially scaffolded behaviour.

Language, narrative, and ritual are not decorative—they are alignment technologies. Boyd and Richerson (1985) formalised cultural evolution as an adaptive amplifier: social learning filters noise, retains effective heuristics, and accelerates recalibration. Henrich and McElreath (2003) show that under high volatility, social transmission systematically outperforms isolated inference. Axelrod (1984), Nowak and Sigmund (1998), and Bowles and Gintis (2011) converge on the principle that sustained cooperation emerges from feedback-enabled communication systems, norm enforcement, and memory.

Cognitive scaffolding matters. Tomasello (1999) and Heyes (2018) show that joint intentionality and shared inferential models enable coordinated pursuit of strategic outcomes. Empirical evidence across species—from fish schools (Ward et al. 2008) to ant colonies (Franks et al. 2002) to human organisations (March 1991; Weick 1995)—confirms that distributed agents outperform isolated ones when collective inference is reinforced by synchronised communication and cultural encoding.

Strategic coherence scales not by dictation, but by shared inference architectures. Communication and culture bind fragmentary cognition into distributed strategic systems capable of sensing, adapting, and acting under drift. Strategy, at scale, is culture operationalised: the recursive alignment of local inference into system-wide foresight.

K. From Empirical Ground to Elemental Grammar

Sections 4.1 through 4.9 have traced the conditions under which strategy emerges as a distinct, necessary, and empirically observable phenomenon. What they reveal is not a singular essence but a constellation of interlocking properties—each derived from a universal constraint, each indispensable to any system attempting to persist and act under uncertainty. This section does not offer a definition. It establishes the full inventory of definitional elements from which Chapter 5 will construct a universal, testable grammar of strategy—and the principles that Chapter 6 will

operationalise into the 18 Fundamental Laws.

Definitional Elements: The Irreducible Properties of Strategy

These are not arbitrary attributes. Each is derived from first principles, observed across biological, artificial, cognitive, and institutional systems. A system that lacks any one of them cannot be said to exhibit strategic behaviour. These are the necessary conditions of strategy:

- **Bounded, Entropic Information**: Strategy exists because information is physical, retrospective, decaying, and incomplete. Systems act strategically because they cannot access full knowledge of their environment. (Shannon 1948; Landauer 1961; Jaynes 2003; Penrose 1989)
- **Probabilistic Futures and Superposition**: The future is not fully knowable; it exists as a structured superposition of probabilities. Strategic action collapses uncertain possibilities into realised outcomes. (Heisenberg 1927; Schrödinger 1935; Busemeyer & Bruza 2012)
- **Recursive Inference and Knowledge Construction:** Strategy requires transforming past signals into beliefs about future states. This is a recursive, inferential process—knowledge is not archived fact, but provisional belief under entropy. (Campbell 1974; Plotkin 1994; Jaynes 2003)
- **Complex-Adaptive Agency**: Strategy is enacted by agents capable of recursive learning, feedback integration, and heuristic rule adjustment. These agents are complex-adaptive systems. (Holland 1995; Gell-Mann 1994; Mitchell 2009)
- **Environmental Drift and Selective Pressure**: Strategy exists within a dynamic, entropic environment. Systems face constant erosion, and only those that produce coherence faster than it degrades persist. (Van Valen 1973; Darwin 1859; Holling 2001)
- **Recursive Feedback and the Dancing Landscape**: Every strategic action reshapes the terrain it acts upon. Advantage is not fixed—it drifts. Strategy must continuously recalibrate under systemic flux. (Kauffman 1993; Thompson 2005; Gould & Eldredge 1977)
- **Coherence as Structural Amplifier:** Coherence is not decoration but architecture. It synchronises local inference across agents, enabling systems to resist drift and act in unified fashion. (Uhl-Bien & Marion 2008; Boisot 1998; Henrich 2015)
- **Formulation, Cultivation:** Execution Strategic behaviour unfolds across three interlinked phases: formulation (intent and belief), cultivation (capability and coherence), and execution (action and collapse). (Porter 1996; Rumelt 2011; Mintzberg 1994)
- **Purposeful Design and Strategic Intent:** Strategic agents pursue not only survival but abstract goals. Strategic action is purposeful orchestration—not reactive drift. Design and intent are integral. (Dennett 1995; Tomasello 1999; Henrich 2015)
- **Communication, Culture, and Collective Inference:** Strategy scales through synchronised collective inference. Language, narrative, ritual, and culture act as scaffolding for system-wide coherence. (Boyd & Richerson 1985; Weick 1995; Tomasello 1999)

Principles to Be Operationalised: The Foundations of Strategic Law

From these elements, we extract the operational principles that Chapter 6 will render into laws. Each principle is observable, falsifiable, and generalisable across contexts:

- The Information-Action Cycle: Strategy requires a complete loop—observe, infer, decide, communicate, act.
- Entropy as Constraint: All information degrades; all action faces drift.
- Knowledge as Probabilistic Inference: Belief must be recursively updated under uncertainty.
- Superposition Collapse: Strategic action collapses futures—before certainty is available.
- Complexity and Emergence: Advantage arises from recursive adaptation, not control.
- Subjective Landscape: The strategic environment is perceived differently depending on inferential architecture.
- Selective Pressure as Arbiter: Strategy is tested by persistence, not elegance.
- Recursive Recalibration: Only systems that continuously realign persist in volatile contexts.
- Distributed Coherence: Culture, communication, and norms synchronise decentralised inference.
- Design and Intent: Strategic entities shape, not merely respond to, their environments.

These principles will be restated as formal, universal laws of strategy in Chapter 6—each providing a falsifiable condition for the presence, absence, or decay of strategic behaviour.

The Canonical Fragmentation—and the Need for Synthesis

The strategic canon has long circled these insights without systematising them. Clausewitz illuminated uncertainty but left coherence unstated. Porter addressed deliberate structure but assumed stable terrain. Mintzberg captured emergence but never constructed grammar. Rumelt gave us diagnosis and coherence, but not recursion. Freedman surveyed the plurality, yet no unified theory stood.

Across disciplines—biology, artificial intelligence, cognitive science, and cultural evolution—the elements are scattered. What QCEA provides is not metaphor, but mechanism: a universal grammar of strategy, grounded in entropy, recursion, and adaptation.

Transition to Definition and Law

The task now is twofold: In Chapter 5, to synthesise these elements into a single, formal definition of strategy that holds across domains. In Chapter 6, to derive from this definition the 18 testable, universal Laws of Strategy—each an operational articulation of how systems persist, adapt, and construct advantage under uncertainty.

What follows is not a philosophical claim. It is a scientific proposition: that strategy, like life, survives by collapsing entropy into coherence. What QCEA offers is the machinery by which that process becomes visible, definable, and operational.

VI. THE FORMAL DEFINITION OF WHAT STRATEGY IS

Having derived each element from empirical constraint and canonical fragment, the QCEA paradigm holds that strategy cannot be defined by choice or tradition alone. It must stand as a phenomenon that persists only where drift tests it and coherence outpaces entropy.

Strategy is a recursive, complex-adaptive process enacted by a strategic entity — itself a complex-adaptive system — operating under conditions of selective pressure within a dynamic, entropic, and emergent dancing landscape of interdependent superpositions and systemic flux.

It comprises formulation, cultivation, and execution — phases that recursively shape and are shaped by the environment. Strategic action uses knowledge to interpret retrospective, entropic information, enabling inferential decisions and the deliberate allocation of finite resources to collapse advantageous superpositions into realised outcomes.

Strategy is not merely a response to complexity — it is the means by which complexity is shaped, coherence is imposed, and agency is exercised through purposeful, knowledge-driven intervention aimed at amplifying favourable outcomes towards an attractor state as defined by the strategic intent.

In its totality, strategy is the ceaseless orchestration of these elements, aligning actions with objectives, transmuting uncertainty into advantage.

The strategy, *in contrast, is the articulated design that includes the strategic logic and a clear set of heuristics that govern action as guiding principles — it is also an evolving portfolio of decisions, priorities, and interventions that manifests this process. If strategy is the process, the strategy is its expressed intent and logic.*

This holds wherever systems persist by collapsing uncertainty into advantage — molecular, cultural, institutional, or artificial. When these elements fail to align, coherence decays, advantage dissolves, and strategy ceases to exist.

What follows is its test: the Laws that bind each condition as an operational invariant, or else expose it to drift — as all strategy must be.

The Recursive Triad: Formulation, Cultivation, Execution

The QCEA paradigm asserts that strategy does not reside in any single action, plan, or outcome. It is a recursive cycle that persists only where three phases remain dynamically aligned under drift.

Formulation is the articulation of intent. It is the provisional inference that maps retrospective, entropic signals onto possible futures. In this phase, justified belief structures — knowledge — shape what advantage means in a given landscape.

Cultivation is the design and maintenance of coherence. It is the work by which a strategic entity aligns its internal capabilities, structures, and norms to survive drift, amplify feedback, and sustain adaptive advantage. Cultivation ensures that local actions do not fragment into entropy but bind into emergent resilience.

Execution is the collapse of superpositional possibility into realised outcomes. It is not merely implementation; it is the decisive orchestration that converts probabilistic futures into selective advantage — continually reshaping the landscape it navigates.

These phases do not proceed linearly. They operate as a recursive triad: each informs and constrains the others. When drift erodes coherence, cultivation must recalibrate. When the environment shifts, formulation must revise intent. Execution tests and refines all assumptions. Strategy persists — or fails — to the extent that this cycle outruns entropy's pull.

TABLE V.

Definitional Clause	**Derived From**
Strategy is a recursive, complex-adaptive process enacted by a strategic entity...	Complex-adaptive agency; Recursive inference and knowledge construction (V–C, V–D)
...operating under conditions of selective pressure within a dynamic, entropic, and emergent dancing landscape...	Environmental drift and selective pressure; Recursive feedback and the dancing landscape (V–E, V–F)
...It comprises formulation, cultivation, and execution...	Formulation, Cultivation, Execution (V–H)
...Strategic action uses knowledge to interpret retrospective, entropic information...	Bounded, entropic information; Recursive inference and knowledge construction (V–A, V–D)
...enabling inferential decisions and the deliberate allocation of finite resources to collapse advantageous superpositions into realised outcomes.	Probabilistic futures and superposition; Knowledge as inference; Superposition collapse (V–B, V–D)
Strategy is not merely a response to complexity—it is the means by which complexity is shaped...	Coherence as structural amplifier; Complexity and emergence; Design and intent (V–G, V–I)
...coherence is imposed, and agency is exercised through purposeful, knowledge-driven intervention aimed at amplifying favourable outcomes...	Design, purpose, strategic intent; Communication, culture, and collective inference (V–I, V–J)
...In its totality, strategy is the ceaseless orchestration of these elements...	All ten definitional elements and principles cohere here: recursive orchestration, coherence, purpose, inference, adaptation, and collapse (V–A to V–J)

VII. IMPLICATIONS & OPERATIONALISATION THROUGH LAWS

If QCEA constitutes a generalisable, falsifiable theory of strategy, then strategy theory must evolve—from heuristic collage to coherent constitution. Three implications follow.

First, we have a definitional anchor. Strategy is no longer a euphemism for ambition or a cipher for power. It is a recursive, adaptive process constrained by entropy—defined not by metaphor, but by the logic of inference under uncertainty.

Second, this definition enables a rigorous research programme. Strategy may now be tested—structurally, not just narratively. Competing approaches can be compared by how they reduce epistemic lag, cultivate coherence, and collapse uncertainty into realised advantage.

Third, the paradigm is operational. QCEA does not merely describe; it constrains. Its elements can be simulated, tested, and refined under drift. From this foundation arise 18 Strategic Laws—each a formal invariant, binding information, inference, and action into the grammar of strategic persistence.

Strategic Law 1: The Law of the Information-Action Cycle

All strategic reasoning operates through a recursive information-action cycle—Observe → Infer → Decide → Communicate → Act. Any entity incapable of completing this cycle is not engaging in strategy but is merely reacting passively to external forces.

Strategic Law 2: The Law of Strategic Systems

Only a complex-adaptive system—one capable of processing retrospective, entropic, and incomplete information, inferring probabilistic future states, and adapting its behavior—can

execute strategy. Systems that lack the ability to process information and recalibrate their actions in response to emerging realities cannot engage in strategy; they can only follow predefined plans.

Strategic Law 3: The Law of Quantum, Retrospective, and Entropic Information

All information is inherently retrospective, entropic, and probabilistically constrained. It describes past states rather than present realities, its fidelity degrades over time due to systemic uncertainty, and it is bounded by quantum observational constraints—ensuring that no signal is ever complete. As a result, strategy is necessarily probabilistic, requiring continuous inference, recalibration, and action under conditions of irreducible uncertainty.

Strategic Law 4: The Law of Inference Through Knowledge

The future can only be inferred by processing retrospective and entropic information through a Bayesian knowledge framework—structured, probabilistically updated beliefs about the implications of information for future states. The accuracy of this inference is constrained by three factors:

- *Information Fidelity: The precision and reliability of incoming data.*
- *Knowledge Framework Quality: The veridicality, epistemic reliability, and coherence of the system interpreting information.*
- *Probabilistic Constraints: The inherent limits of forecasting in complex, dynamic environments.*

Since perfect foresight is unattainable, strategic entities must operate on continuously updated probabilistic beliefs rather than fixed assumptions.

Strategic Law 5: The Law of Entropic Superposition

The future exists in three forms: deterministic trajectories, probabilistic superpositions, and random chaos. Strategy applies only to probabilistic superpositions—where uncertainty is structured, and inference and action can tilt outcomes. Deterministic futures require only planning, while random chaos is beyond strategic influence.

Strategic Law 6: The First Law of Strategic Subjectivity

The future is not an objective reality—it is a construct shaped by the observer's knowledge frameworks. The structure of uncertainty is determined by the sophistication of strategic perception.

- *Two entities can face identical conditions but perceive entirely different landscapes based on their ability to recognise patterns.*
- *Strategic superiority is not only about responding to future events but defining what the future even means.*
- *Entities that fail to update and refine their classification of uncertainty are outpaced by those that do.*
- *The ability to manipulate an adversary's perception of uncertainty is itself a strategic weapon.*

Strategic Law 7: The Law of the Dancing Landscape

Strategy exists only within complex-adaptive systems where the future unfolds as a structured, probabilistic superposition. These conditions ensure that the future does not follow a single trajectory but emerges as a dynamic, interconnected lattice of probabilities—continuously reshaped by historical dependencies, emergent interactions, and systemic interconnections. The evolution of any strategic context is driven by both endogenous (internal) and exogenous (external) forces, which interact to collapse, reform, and reconfigure probability distributions over time.

Strategic Law 8: The Second Law of Strategic Subjectivity

The perceived rate, magnitude, and jerk of a dancing strategic landscape are subjective, determined by the observer's information-action cycle.

- *Rate (how fast the landscape appears to move) is directly proportional to lag in the strategic organism's information-action cycle.*
- *Magnitude (how extreme the changes appear) is directly proportional to error in the strategic organism's information-action cycle.*
- *Jerk (how violently and unpredictably the landscape seems to shift) is directly proportional to both lag and error.*

Thus, the more slowly and inaccurately an entity processes and acts on information, the more unstable and chaotic the environment appears—regardless of its objective reality.

Strategic Law 9: The Law of Adaptive Need and Strategic Balance

The adaptive need of a strategic organism is directly proportional to its total entropy (including lag and error). In low-entropy environments, strategic decisions emerge from structured, algorithmic processes, while in high-entropy environments, decision-making shifts toward heuristic-driven approaches. As entropy increases, strategic organisms must prioritise adaptation over optimisation, as algorithmic precision becomes unviable. This phenomenon is localised within different parts of a strategic system, creating zones of determinism and zones of adaptive uncertainty.

Strategic Law 10: The Law of Strategic Intent

Strategic intent is the necessary condition for strategy, defining the pursuit of advantage within an evolving attractor state. As long as strategic intent persists, strategy exists; when intent ceases, strategy terminates.

In finite contexts (e.g., games, military campaigns, targeted market entry), strategy operates until the terminal objective is achieved, after which its function dissolves.

In open-ended contexts (e.g., corporate competition, geopolitical positioning), goals are transient—mere waypoints in an ongoing process of adaptation and opportunity exploitation.

Strategic organisms that fail to realign their intent with shifting attractor states will experience displacement or loss of advantage.

Strategic Law 11: The Law of Cultivating a Favourable Context

Strategic organisms do not operate within static conditions but within evolving selective pressures. The pursuit of advantage is not about achieving fixed objectives but about continuously reshaping the environment to align with emerging attractor states. Strategic viability depends on the capacity to influence or adapt to shifting pressures; failure to do so results in displacement by more adaptive competitors.

Strategic Law 12: The Law of Coherence and Pattern Exploitation

Strategic success derives not merely from recognising patterns but from the deliberate engineering of coherence—amplifying advantageous configurations and attenuating disruptive ones. Entities that align their actions with coherent objectives generate self-reinforcing strategic momentum. The capacity to discern, enhance, and sustain emergent patterns distinguishes high-performing strategic actors from those entrapped in reactive drift. Misalignment between strategic aims and systemic forces results in entropy and wasted energy. Strategy, then, is not the art of passive adaptation, but the discipline of shaping coherence in order to construct advantage.

Strategic Law 13: The Law of Timing, Sequence, and Path Dependence

The trajectory of any strategic system is shaped by its initial conditions, the timing and sequence of strategic actions, and the emergent constraints imposed by its evolving landscape. Once a strategic path is taken, future options are either enabled or constrained by prior actions, making reversibility limited and strategic flexibility path-dependent. The probability distribution of future superpositions is continuously shaped by prior decisions and external environmental interactions.

Strategic Law 14: The Law of Goal-Maintaining vs. Goal-Achieving Behaviour

Strategic organisms must exhibit goal-maintaining behaviour at the level of the organism, directed by strategic intent, which is an emergent property of the collective, actor-level goal-achieving behaviour of the actors that make up the strategic organism. In strategic organisms, individual actors will tend to default to goal-achieving behaviours, even when the overall system requires goal-maintaining behaviours. These local goal-achieving behaviours are typically governed by simple rules—heuristics or algorithms—optimised for immediate feedback loops rather than systemic coherence. When unregulated, such decentralised optimisations can override higher-order strategic needs, leading to systemic misalignment, inefficiency, or collapse.

Strategic Law 15: The Law of Strategic Enablers

Strategy is an emergent phenomenon of complex-adaptive systems, dependent on the cooperation and coherence of actors within the strategic organism. This coherence is sustained by strategic enablers—beliefs, norms, culture, values, knowledge systems, and behaviours—that structure the information-action cycle, determining whether actors align with shared strategic intent. Strategic enablers must be actively cultivated, continuously adapted, and periodically restructured to sustain strategic coherence over time. A strategic entity that fails to cultivate, adapt, and align its enablers does not merely lose efficiency—it ceases to function as a strategic organism.

Strategic Law 16: The Law of Survival as a Necessary Condition

The survival of the strategic organism is a necessary condition for the execution of the strategy. While strategy can prioritize objectives detached from survival, selective pressures continue to operate. A strategic organism that optimizes for objectives at the expense of its own survival will eventually lose strategic viability and risk self-termination. Strategic execution is constrained by survival, even when survival is not the primary goal.

Strategic Law 17: The Law of Resolution and Unit of Analysis

The Law of Resolution and Unit of Analysis Strategic effectiveness fundamentally depends on selecting the appropriate level of resolution to perceive, interpret, and act upon complex systemic patterns. Choosing too broad or too detailed a lens—macro blindness or micro-paralysis—results in inefficiencies, flawed decisions, and systemic failures. True strategic acumen lies in dynamically shifting between resolutions, applying precise analytical disciplines tailored to each level.

Strategic Law 18: The Law of Strategic Logic

The coherence and effectiveness of strategy are governed by the dynamic equilibrium between two interdependent forces: how an entity positions itself in relation to the external environment (exogenous logic), and how it internally structures itself to act upon that positioning (endogenous logic). This dual structure—of looking outward while cultivating inward—forms the strategic logic of the entity. It is not a static doctrine but a continuously recalibrated mechanism designed to ensure that intent is not merely declared but executed coherently, adaptively, and persistently.

The eighteen laws do not stand as a loose enumeration but as an integrated system, their architecture echoing the recursive nature of strategy itself. They are arranged into three foundational strata, each building upon and constraining the next.

Foundational Laws (1–5): The Ontological Substrate

These laws specify the conditions that must exist for strategy to be possible in any domain. Law 1 (Information–Action Cycle) delineates the minimal recursive loop required for strategic behaviour. Law 2 (Strategic Systems) defines the nature of the entities capable of executing this loop. Laws 3–5 impose the information-theoretic constraints within which all strategic activity unfolds: retrospective entropy, the construction of inferential knowledge, and the probabilistic superposition of possible futures. Absent these conditions, strategy collapses into incoherence.

Interpretive Laws (6–9): The Subjective Lens

Resting upon this ontological base, the interpretive laws describe how strategic entities

perceive and frame their realities. Laws 6 and 8 (Strategic Subjectivity) establish that the strategic landscape is never a neutral tableau but is constructed through the observer's own inferential architecture. Law 7 (Dancing Landscape) captures the inherently dynamic, shifting nature of these environments, while Law 9 (Adaptive Need) links the entropy of the environment to the requisite intensity and precision of strategic response. Together, these laws explain why entities sharing the same external conditions nonetheless inhabit fundamentally different strategic worlds.

Operational Laws (10–18): The Executive Architecture

The final tier governs the mechanics of action. Laws 10–13 address the inner workings of strategic execution: the formation of intent, the cultivation of context, the engineering of coherence, and the orchestration of time. Laws 14–15 extend this to the organisational problem of sustaining coherence across distributed actors and decision layers. Law 16 enshrines survival as the ultimate boundary condition, while Laws 17–18 address the meta-challenges of analytical resolution and strategic logic. These laws translate perception into purposeful action within the constraints defined by the preceding layers.

This tripartite structure is inherently recursive. Operational moves generate fresh information that re-enters the foundational cycle, shifting interpretive frames and demanding new courses of action. Each layer thus both enables and constrains the others in a ceaseless dynamic of strategic becoming—a structure as much lived as it is theorised.

Each law articulates a modular invariant enabling strategy to function across dynamic contexts and varying resolutions of uncertainty. While not elaborated here, they constitute the practical scaffolding by which QCEA may be enacted, refined, and diagnostically interrogated.

Together, these Laws formalise the full architecture by which any system — molecular, biological, institutional, or artificial — must convert fragmentary, entropic information into sustained advantage. Each stands as a falsifiable condition by which the theory endures or fails under selective pressure, as strategy itself always must.

Finally, the paradigm invites integration. Strategy need no longer sit uneasily atop imported disciplines. It may now offer a reciprocal grammar through which cognition, computation, evolution, and agency can be understood across contexts. In doing so, strategy may finally become what it has long pretended to be: a master discipline—not by imperialism, but by synthesis.

The work ahead is vast. But it is now possible. The QCEA paradigm does not end the conversation; it reorients it. From metaphor to mechanism. From plan to process. From artefact to architecture.

That, perhaps, is strategy.

TABLE VI.

Strategic Law	**Rooted in Definitional Elements**
Law 1: The Information-Action Cycle	Recursive inference; Complex-adaptive agency (V–C, V–D)
Law 2: Strategic Systems	Complex-adaptive agency; Feedback and adaptation (V–C, V–F)
Law 3: Quantum, Retrospective, and Entropic Information	Bounded, entropic information; Probabilistic futures (V–A, V–B)
Law 4: Inference Through Knowledge	Recursive inference; Knowledge construction (V–D)
Law 5: Entropic Superposition	Probabilistic futures and collapse dynamics (V–B)
Law 6: First Law of Strategic Subjectivity	Superposition; Inferential architecture (V–B, V–D)
Law 7: Dancing Landscape	Recursive feedback; Environmental flux (V–F)
Law 8: Second Law of Strategic Subjectivity	Feedback loop lag and error; Complex-adaptive perception (V–F, V–D)
Law 9: Adaptive Need and Strategic Balance	Entropy; Localised adaptation (V–A, V–F)
Law 10: Strategic Intent	Purposeful design; Complex-adaptive orchestration (V–I, V–C)

Law 11: Cultivating a Favourable Context	Environmental drift; Strategic influence (V–E, V–I)
Law 12: Coherence and Pattern Exploitation	Coherence as amplifier; Complexity (V–G, V–F)
Law 13: Timing, Sequence, and Path Dependence	Recursive feedback; Environmental emergence (V–F, V–E)
Law 14: Goal-Maintaining vs. Goal-Achieving Behaviour	Strategic phases; Distributed agency (V–H, V–G)
Law 15: Strategic Enablers	Culture; Communication; Collective inference (V–J, V–G)
Law 16: Survival as Necessary Condition	Selective pressure; Strategic viability (V–E, V–I)
Law 17: Resolution and Unit of Analysis	Probabilistic inference; Recursive architecture (V–B, V–D)
Law 18: Strategic Logic	Endogenous/exogenous coherence; Strategic orchestration (V–F, V–G, V–I)

VIII. UNIVERSAL IMPLICATION AND CONCLUSION

Strategy, as clarified through the QCEA paradigm, is neither a managerial slogan nor a mere artefact of war, politics, or commerce. It is a universal phenomenon: a recursive, complex-adaptive process by which systems harness fragmentary, entropic information to collapse uncertainty into advantage under selective pressure.

Sections 4.1 through 4.9 demonstrate that this orchestration requires irreducible elements: the information-action cycle, complex-adaptive agency, knowledge-driven inference, recursive feedback, emergent coherence, purposeful design, collective alignment, and the ability to reshape a shifting terrain. Each element is not optional. Each is surfaced from first principles—physics, evolutionary epistemology, complexity science, comparative cognition—and bound together into a single operational architecture.

Where the canonical tradition illuminated fragments, the QCEA paradigm binds them into an integrated mechanism. Where prior frameworks offered heuristics, it offers testable invariants. The eighteen Strategic Laws formalise this claim: each stands as an empirical condition that any system must satisfy to enact strategy—or else drift dissolves its coherence, as all non-strategic plans must.

The result is a theory not constrained by domain or discipline. Whether applied to a bacterial colony, a multinational firm, an insurgency, or an artificial agent, the test remains the same: does the system orchestrate information, inference, and coherent action faster than entropy can erode its advantage? If so, strategy persists. If not, it collapses—as it must.

Thus, the QCEA paradigm closes the definitional fracture that has long made strategy plural, partial, and unfalsifiable. It does not discard the canonical insights; it absorbs and operationalises them. From metaphor to mechanism, from plan to process, from artefact to architecture: strategy stands defined not by what we wish it to be, but by what survives when drift tests coherence under complexity.

What remains is the test—empirical, recursive, and open to refinement—as all strategy must be.

With this definition and its Laws established, the implications are both theoretical and practical. Academics now hold a universal, falsifiable scaffold for testing strategy as a phenomenon across domains—without defaulting to borrowed metaphor. Practitioners gain a complete framework for formulating, cultivating, and evaluating strategies under real conditions of drift, emergence, and reflexivity. This equips them to diagnose where coherence is sustained, where inferential loops are drifting, and where enablers must be cultivated to outpace entropy. Crucially, it enables lessons to travel across disciplines—by abstraction to core principles, not vague analogy. The cumulative shortfall revealed in the canonical record is resolved here:

- Conceptual clarity — the phenomenon is defined ontologically, not just metaphorically.
- Explanatory scope — the framework holds wherever systems persist by collapsing

uncertainty into advantage.

- Diagnostic power — each Law provides an empirical handle to detect coherence and drift.
- Generativity — the framework is modular, recombinable, and open to simulation.
- Falsifiability — each condition is a testable invariant that drifts will erode if coherence is not sustained.

This does not dictate what strategy must be—it operationalises what strategy does when it works.

A. Boundary Conditions

The QCEA paradigm applies wherever systems operate under bounded information, selective pressure, and environmental drift—conditions that render persistent advantage contingent upon the orchestration of information, inference, and action. Its scope is universal in abstraction, yet bounded in application. It does not apply in:

- Deterministic systems where outcomes are fully knowable and fixed.
- Random or chaotic systems where outcomes are aleatory and unpatterned (e.g., roulette).
- Passive systems lacking feedback, inference, or agency.
- Purely reactive models that neither forecast nor recalibrate.

The contrast is instructive: while roulette admits no strategy beyond superstition, poker demands recursive inference, belief revision, and probabilistic calibration—QCEA's exact conditions.

B. Operationalisation and Future Research

This paper deliberately refrains from full empirical operationalisation of the eighteen Strategic Laws. Each law implies a family of observable, falsifiable conditions. Translating these into robust research instruments—capable of capturing recursive inference, coherence decay, and strategic collapse—remains the primary task for future inquiry.

We identify four interlocking streams of empirical development:

1. Measurement-Protocol Development

Develop standardised instruments to detect and diagnose the operational signatures of each law.

- Design entropy analytics, signal-coherence maps, and structured surveys to capture inferential drift, coherence formation, and entropy navigation.
- Validate construct and predictive validity through pilot studies in contrasting contexts (e.g., software start-ups versus military command teams).
- Extend instrumentation into real-time trace mapping of decision loops and post-hoc coherence analytics.

Here, critical realist evaluation studies (Thomas, 2010) offer a pathway for maintaining ontological depth while engaging observable organisational patterns. Thomas' operationalisation of critical realism in management science ensures that metrics preserve structural causality rather than default to correlational heuristics.

2. Longitudinal and Simulation Studies

Build dynamic models to test strategic persistence under temporal and environmental drift.

- Conduct process-based longitudinal ethnographies (Chia, 2008) to map the lived rhythm of inference, disruption, and recovery.
- Simulate collapse-recovery cycles under controlled volatility using agent-based or system-dynamics models.
- Stress-test law interactions under high-entropy conditions to model strategic failure and adaptation.

Chia's methodology provides not only empirical texture but temporal granularity—making visible the recursive recalibration that underpins QCEA's architecture.

3. Strategic Artefact and Documentary Analysis

Analyse strategic texts, dashboards, planning artefacts, and meeting protocols as objects of inference.

- Use documentary analysis (Ferraris, 2013) to trace how coherence is externalised, institutionalised, and degraded over time.
- Map the material-discursive interplay between strategy-as-claimed and strategy-as-enacted.
- Develop coherence scoring methods for strategic artefacts, assessing their ability to encode, preserve, and adapt inference.
- Ferraris' hermeneutic reading of organisational documents aligns with Laws 10–12, where coherence must be both perceived and embedded.

4. Cross-Domain Generalisability Tests

Empirically assess whether strategy—as defined—travels across domains.

- Apply protocols to ecological systems, insurgent campaigns, regulatory agencies, and autonomous systems.
- Compare drift-resilience, inferential precision, and coherence durability.
- Delimit where strategy, as recursive coherence under entropy, ceases to apply.
- Object-oriented ontology, particularly in organisational studies (Harman, 2018), assists here: treating strategy as a system-level object whose causal profile must be traced through relations and manifestations, not simply observed behaviours.

Executing this agenda would elevate QCEA from falsifiable theory to cumulative science. It would convert the paradigm from ontological grammar to empirical scaffold—capable not only of diagnosing strategic coherence but of predicting its decay, recovery, and collapse under real-world volatility.

Clarifying Note on Indicators.

Detailed examples of empirical indicators for each of the eighteen Laws appear in the Appendix. These are provided illustratively; a full methodological exposition is beyond this article's scope and will be developed in subsequent empirical work.

Testing Pathways:

- Agent-based modelling: Simulating entities with and without inferential depth across volatile landscapes.
- Organisational signal tracing: Mapping how information and belief update through decision layers.
- High-frequency ethnographic capture: Observing strategic recalibration in-situ to surface the presence (or absence) of recursive inference.

Practioner Implications:

- Do not confuse goals with strategy. Begin with a clear theory of how outcomes emerge—not just what outcomes are desired.
- Model the environment as a system of constraints and feedbacks. A wish is not a plan; simulate cause, not just intention.
- Diagnose coherence gaps. Misalignments in timing, narrative, or resource flow signal where drift is eroding advantage.
- Adapt structure to scale. Small teams and large bureaucracies require different inference architectures.
- Treat complexity as a navigational variable. Calibrate how much emergence can be forecast versus absorbed.
- Build inference capability, not just execution muscle. Strategic action without learning mechanisms is erosion-in-waiting.
- Track epistemic lag. Late learning is lost advantage.
- Distribute strategic sensing. The edge often sees what the centre cannot.
- Use simulations to test collapse points. Do not wait for drift to surface through failure.
- Redesign feedback loops. Make them frequent, falsifiable, and capable of informing the next inferential cycle.
- Measure not just what's done, but what's believed. Strategy decays when beliefs diverge unobserved.
- Create redundancy in coherence structures. If one loop fails, another must hold.
- Codify what is held stable and what must adapt. Cultivate intentional stability—not inertia.
- Ensure recursive update. Strategy is not a document; it is a loop.
- Differentiate tactical agility from strategic drift. Not all change is progress.
- Use narrative as signal compression. Well-crafted stories align inference across actors.
- Design for coherence, not compliance. Systems that align from within outperform those that enforce from above.
- Preserve strategic viability as a precondition. Pursuit of non-survival goals only endures if coherence is sustained.
- Frame problems in probabilistic terms—uncertainty cannot be eliminated, but it can be collapsed into informed action through better framing and belief revision.
- Surface multiple hypotheses before committing—strategy begins not with solutions, but with competing inferences that must be tested under drift.
- Distinguish signal from noise—rely on information structures that reduce entropy rather than accumulate irrelevant data.
- Avoid premature convergence—delay irreversible choices until sufficient information justifies collapse into action.
- Design feedback loops that refine belief—strategy is not about control, but about adaptive recalibration under uncertainty.
- Model the system, not just the outcome—advantage emerges from intervening in the structure, not just chasing metrics.
- Cultivate coherence across functions—aligned beliefs, actions, and temporal horizons are more resilient than high-performing silos.
- Map the topography of influence—understand how local moves deform the strategic terrain

for others and yourself.

- Build redundancy into your decision logic—fragile strategies collapse under variance; robust ones anticipate failure.
- Treat time as entropic, not linear—past performance decays; future relevance must be continuously re-inferred.
- Select for optionality, not efficiency—adaptive capacity outperforms static optimisation under volatility.
- Model competitors as inferential agents—assume others are collapsing their own superpositions, not just reacting to you.
- Design systems that learn at multiple levels—tactical lessons must feed strategic recalibration, not just operational tweaks.
- Monitor coherence drift explicitly—loss of strategic alignment begins silently; diagnostic instruments must surface it early.
- Structure decisions as recursive experiments—each choice should teach, not merely conclude.
- Cultivate the enablers of future inference—invest in talent, culture, and memory as strategic scaffolding.
- Align incentives to strategic belief states—strategy cannot persist where reward undermines inferred direction.
- Accept that strategy is never final—it is a continuous act of navigation, re-inference, and coherence renewal under drift.

C. Limitations and Future Research

The QCEA paradigm is not a behavioural theory. It does not model preference formation, affect, or decision heuristics at the individual level. Instead, it formalises the conditions under which strategic agency must operate if it is to persist. Accordingly:

- Not psychologically grounded: It models what systems must do, not what individuals typically do.
- Not yet simulated: The recursive machinery is formalised but not yet instrumented at scale.
- It does not simulate bounded rationality directly, but rather prescribes the conditions that strategic behaviour must fulfil under entropy and drift.
- It assumes recursive inference and selective feedback as foundational—thus excluding fully chaotic, random, or deterministic environments where strategic adaptation is either irrelevant or redundant.
- The formal framework remains theoretical. Instrumentation, simulation, and ethnographic verification are necessary to make the Laws empirically tractable.

D. Limitations and Future Research

While this work grounds the ontology of strategy through philosophical analysis, it necessarily inaugurates a programme of empirical investigation. The QCEA framework is not a closed metaphysical system, but a generative architecture that yields testable propositions—what Merton (1968) termed a *theory of the middle range*: abstract enough to unify diverse phenomena, yet sufficiently concrete to structure empirical inquiry.

The framework generates precise theoretical expectations. It predicts, for instance, that

strategic failure will correlate with measurable degradation in the completion times of the information–action cycle (*Law 1*). This invites longitudinal studies examining the relationship between decision latency and performance outcomes—a quantifiable trace of entropic drag.

The tripartite distinction between quantum, retrospective, and entropic information (*Law 3*) implies that organisations characterised by high Shannon entropy in their decision environments will require more sophisticated inference architectures to sustain coherence and efficacy. This is a testable hypothesis, amenable to investigation through information audits and complexity-informed diagnostics.

The laws of strategic subjectivity (*Laws 6 and 8*) yield particularly rich empirical entailments. They predict that two organisations, even when embedded in identical external conditions, will perceive divergent strategic landscapes due to differences in their information-processing architectures. Comparative studies may therefore explore how firms within the same sector construct qualitatively distinct mental models of competition and opportunity—with corresponding divergence in strategic choice and outcome.

Perhaps most provocatively, the framework asserts that *coherence* (*Laws 12 and 15*) operates as a measurable system property mediating between declared intent and realised consequence. This invites the development of novel metrics for organisational coherence—potentially via network analysis of information flows, semantic mapping of strategic communications, or signal-pattern detection in resource allocation regimes. Coherence, in this view, is not a rhetorical virtue but an empirical variable.

The QCEA framework thus offers a structured epistemic scaffolding for future research. Scholars may choose to operationalise individual laws, examine boundary conditions through comparative or historical analysis, or reinterrogate existing strategic datasets through an information-theoretic lens. In each case, the task is not merely validation but elaboration—testing not only *whether* the framework holds, but *how* its generative logic can illuminate the hidden structure of strategic behaviour.

CONCLUSION

Strategy, thus reconstituted, is no longer a matter of metaphor but of mechanism. The QCEA paradigm renders it intelligible not as a managerial aspiration but as a universal process — recursive, entropic, adaptive. It defines what strategy *is*, not merely what strategists *do*. What follows is no longer rhetorical speculation but an empirical architecture, open to refinement and falsification. If the field is to advance, it must now do so not by collecting further anecdotes, but by testing coherence under drift. The age of strategic poetry ends; the discipline of strategic science begins.

APPENDIX

TABLE VII.

Law	**Empirirical Indicator**	**Measurement Method**
Law 1: Information-Action Cycle	Cycle completion time (hours/days from observation to action)	Track timestamps from environmental signal detection to response implementation
Law 2: Strategic Systems	System responsiveness coefficient (% of external changes triggering internal adjustments)	Monitor environmental changes vs. organizational responses over rolling 6-month periods
Law 3: Quantum/Retrospective/Entropic Information	Information entropy index (Shannon entropy of decision inputs)	Calculate $H = -\Sigma p(i)\log_2 p(i)$ for probability distributions of key variables
Law 4: Inference Through Knowledge	Knowledge utilization ratio (% of available information used in decisions)	Compare information accessed vs. information available in decision processes
Law 5: Entropic Superposition	Option diversity index (number of viable strategic alternatives maintained)	Count and weight strategic options kept open at decision points

Law 6: First Law of Strategic Subjectivity	Forecast error variance (standard deviation of prediction accuracy)	Measure variance in strategic forecasting performance over time
Law 7: Dancing Landscape	Environmental volatility coefficient (rate of change in key market indicators)	Track coefficient of variation in relevant environmental metrics quarterly
Law 8: Second Law of Strategic Subjectivity	Perception-reality gap (% difference between predicted and actual outcomes)	Compare strategic predictions to realized outcomes over 12-month periods
Law 9: Adaptive Need and Strategic Balance	Adaptation frequency ratio (strategic adjustments per environmental change)	Count strategic modifications per significant environmental shift
Law 10 action alignment score (% of stated intentions converted to actions)	Measure conversion rate from strategic declarations to implemented initiatives	
Law 11: Cultivating Favourable Context	Context influence index (% of environmental factors directly shaped by entity)	Assess proportion of controllable vs. uncontrollable environmental variables
Law 12: Coherence and Pattern Exploitation	Strategic coherence coefficient (consistency across strategic elements 0-1 scale)	Evaluate alignment between resources, capabilities, and actions using correlation analysis
Law 13: Timing, Sequence, Path Dependence	Temporal optimization ratio (% of actions taken at optimal timing)	Compare action timing to retrospectively identified optimal windows
Law 14: Goal-Maintaining vs. Goal-Achieving	Goal-behavior classification ratio (% time in maintenance vs. achievement modes)	Track resource allocation between maintaining current position vs. advancing
Law 15: Strategic Enablers	Enabler effectiveness index (performance improvement attributable to culture/communication)	Measure performance changes linked to cultural and communication investments
Law 16: Survival as Necessary Condition	Survival probability score (calculated resilience under stress scenarios)	Use stress-testing methodologies to assess survival likelihood
Law 17: Resolution and Unit of Analysis	Resolution consistency index (coherence across different analytical levels)	Measure correlation of insights across individual/team/organizational levels
Law 18: Strategic Logic	Logic coherence score (internal consistency of strategic reasoning 0-1 scale)	Evaluate logical consistency using formal reasoning frameworks

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